The full correlation range, filtering frontier and volume trade-off at fixed steering centre

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Abstract

For two-qubit states we study the three-setting correlation value $F_3 = \|T\|_{\mathrm{F}}$ of Eq. (3) of Ducuara, Susa and Skrzypczyk (arXiv:2406.03945) as a function of the steering-ellipsoid geometry. An earlier package in this collection determined the exact supremum $G(c)$ of F_3^2 at fixed centre length c, with violation $F_3 > 1$ possible exactly when $c < 2/3$. Here we determine the entire attainable set at fixed centre length c and Bob Bloch length $r < 1$: it is the closed interval $\big[\max\{0, c(1+r)-1\}^2,\ (1-r^2)(3-4c) + (c+(1-c)r)^2\big]$, with separable diagonal minimizers, a classification of the lower endpoint and a trace-norm stability estimate. For canonical states with centre length c and radial support half-width t, the squared correlation $\tau = \|S\|_{\mathrm{F}}^2$ fills $[t^2, 1-c^2+2t^2]$, and the greatest ellipsoid volume at prescribed (c,t,τ) is $(4\pi/3)(t/2)\min\{\tau - t^2,\ 1+2t+2t^2-c^2-\tau\}$, attained at every feasible triple. A canonical input and a Bob filter succeeding with probability at least $p \in [1/2, 1]$ can produce $F_3 > 1$ exactly when $c < 2/[1+p+\sqrt{(1-p)^2+2}]$. We also solve every one-sided filtering orbit exactly, including singular hard cases, and classify the maximizers of $G(c)$ with explicit trace-norm rigidity. Lean 4 with Mathlib proves the central statements for actual complex 4×4 density matrices, Kraus operators and local unitaries; the remaining statements, listed in Appendix C, have conventional proofs.

1 Introduction

Ducuara, Susa and Skrzypczyk [1] relate hidden steering to the steering ellipsoid of Jevtic, Pusey, Jennings and Rudolph [3]. They leave open the function bounding the three-setting correlation value $F_3 = \|T\|_{\mathrm{F}}$ in terms of the centre length c of Alice's steering ellipsoid, and report from random states an apparent cutoff near $c \approx 0.66$ for violation of $F_3 \le 1$. An earlier package in this collection, *Exact three-setting correlation bounds at fixed steering centre* [2], dated 29 October 2026, answered that question: it proved that the least upper bound of F_3^2 at centre length c is $G(c) = 1 + (2-3c)^2/(2-2c-c^2)$ for $c \le 2/3$ and 1 for $c > 2/3$, attained exactly for $c < 2/3$ and $c = 1$, together with the joint upper bound $F_3^2 \le H_c(r)$ at Bob Bloch length r and its attainment along one explicit family. This paper determines the remaining parts of the picture.

The question and the answers in plain terms. In Bloch form a two-qubit state has three ingredients: Alice's local vector $\mathbf{a}$, Bob's local vector $\mathbf{b}$, and the 3×3 matrix T of correlations between spin measurements on the two qubits. The number $F_3 = \|T\|_{\mathrm{F}}$ measures how strongly the spins are correlated along three orthogonal measurement directions; a value above one violates the corresponding three-setting steering inequality. Alice's steering ellipsoid is the set of Bloch vectors to which Bob's measurements can steer Alice's qubit, and its centre and shape constrain T. The earlier package found the largest F_3 compatible with a given centre length. This paper answers three follow-up questions.

- *Which values of F_3 are possible once both the centre length and the length r of Bob's Bloch vector are fixed?* Exactly an interval. Its new lower end says that a long centre together with a strongly polarized Bob forces some correlation, because otherwise the probability of one particular pair of measurement outcomes would be negative.
- *When can a local filter on Bob's side, required to succeed with probability at least p, reveal a hidden violation?* Exactly when the centre length lies below an explicit threshold, which decreases from $2/3$ at $p = 1/2$ to $2 - \sqrt{2}$ at $p = 1$.
- *How large can the steering ellipsoid be when its correlation strength is fixed?* At most an explicit piecewise linear cap in the squared correlation, attained everywhere. The states with the largest correlation do not have the largest volume: the two objectives compete for the same off-diagonal entries of the density matrix.

Main results. For centre length $c \in [0, 1]$ and Bob Bloch length $r \in [0, 1)$:

1. The set of attainable values of F_3^2 is exactly the interval $[L(c, r), H_c(r)]$, $L(c, r) = \max\{0, c(1+r) - 1\}^2$, $H_c(r) = (1 - r^2)(3 - 4c) + (c + (1 - c)r)^2$ (Theorem 5). The lower endpoint has separable diagonal witnesses, every minimizer is locally unitarily equivalent to the witness, and a state with $F_3^2 = L(c, r) + \eta$ lies within squared trace norm $(1 + r^2)\eta$ of an explicit minimizer. At $r = 1$ the only value is c^2.
2. For canonical states (Bob maximally mixed) with centre $c\mathbf{n}$ and radial support half-width $t = \left\|S^{\mathsf{T}}\mathbf{n}\right\|$, the attainable squared correlations form $[t^2, 1 - c^2 + 2t^2]$, for exactly $0 \le t \le 1 - c$ (Theorem 7); the signed determinant fills $[-t((1 + t)^2 - c^2)/4,\ t((1 - t)^2 - c^2)/4]$; and at prescribed (c, t, τ) the greatest $|\det S|$ is $(t/2)\min\{\tau - t^2, 1 + 2t + 2t^2 - c^2 - \tau\}$, attained at every feasible triple (Theorems 11 and 12). The ellipsoid volume is $(4\pi/3)|\det S|$. At $t = 1 - c$ this recovers the fixed-centre volume maximum $(4\pi/3)(1 - c)^2$ of Milne et al. [5].
3. Over canonical inputs of centre length c and Bob Kraus operators with success probability at least $p \in [1/2, 1]$, the greatest F_3^2 of the normalized output is $H_c(\widehat{r})$ with $\widehat{r}$ the clipped vertex of H_c on $[0, (1 - p)/p]$, and violation is possible exactly when $c < 2/[1 + p + \sqrt{(1 - p)^2 + 2}]$ (Theorem 9 and Corollary 10).
4. Every one-sided filtering orbit of a canonical state is solved by an explicit spectral algorithm with at most seven algebraic candidates, including singular and hard cases, with all maximizing effects classified (Sections 10 and 11).
5. The maximizers of $G(c)$ are classified up to local unitaries, with explicit trace-norm stability of the canonical optimizer (Theorems 13 and 15, Appendix A).

What is new, and what is not. The envelope $G(c)$, the threshold $c < 2/3$, the joint upper bound $H_c(r)$ with its attaining family, and the fully entangled fraction bound are results of the earlier package [2]; Sections 4 and 4.1 recall them so that this paper is self-contained. New relative to that package are the lower bound $L(c, r)$ and the complete joint interval, the lower-endpoint classification and stability, the canonical (c, t) interval, the probability-constrained frontier, the complete solution of the one-sided filtering orbits, the classification and trace-norm rigidity of the maximizers, the flat-branch stability exponents, and the signed and three-parameter volume bounds. Classical ingredients are credited where they are used: the steering-ellipsoid and canonical-filtering framework and the maximally obese states [3, 4, 5], the spin-flip method and Bloch-matrix positivity [4, 3, 8], the measurement optimization behind the interpretation of F_3 [6], marginal spectral constraints [7], and trust-region theory for quadratics on a ball [9, 10]. The fixed-centre volume maximum is Theorem 3 of [5]. A bounded literature search did not find the new frontiers; this does not establish priority.

Earlier public record. On 31 October 2026 a check of public records found that the Omega Institute's public repository records a claimed Lean proof of Conjecture 2 of Milne, Jennings, Jevtic and Rudolph [4], the fully entangled fraction bound $f \le 1 - c/2$ at fixed centre, committed on 8 October 2026 (issue #14206) [12], and a proof of their Conjecture 1, a sharp bound on the CHSH value at fixed centre, committed on 14 October 2026 (issue #16558) [13]. The Lean project of this package contains, unchanged from the earlier package, a proof of the fully entangled fraction bound; no priority is claimed for it, and it is not a result of this paper. The CHSH value is a different quantity from F_3. The CHSH dossier also records, as an intermediate lemma proved with the same filtering covariance identity, an upper bound on the CHSH value in terms of both c and Bob's Bloch length; it lists the F_3 centre bound as open. As read on 3 November 2026, neither dossier treats the lower bound or the full interval of F_3^2, the probability frontier, the filtering orbits or the volume bounds. Xing and Yang [11] give volume-based bounds on entanglement and Bell nonlocality through the steering ellipsoid; their full text was not compared with this paper.

Verification. Lean 4.19.0 with Mathlib proves the joint interval, the lower-endpoint equality and trace-norm rigidity, the canonical (c, t) interval, the probability frontier from actual Kraus operators, the complete orbit solution, the classification and rigidity of the maximizers, and the signed and three-parameter volume maxima, for actual complex density matrices; a whole-project audit confirms that every declaration other than compiler-generated auxiliaries depends only on the standard axioms. The flat-branch exponents, the volume-maximizer orbit classification and rank counts, the attainment of every signed determinant, and the physical interpretation of F_3 rest on conventional proofs. Appendix C gives the details.

Organization. Section 2 fixes notation. Section 3 proves a rank-safe geometric inequality and the sharp centre and radial-width envelope, and Section 4 recalls the envelope of the earlier package; everything else builds on these. Sections 5–7 contain the new joint interval, the canonical interval with the probability frontier, and the volume bounds. Sections 8–11 classify the maximizers, quantify their stability and solve the one-sided filtering orbits. Section 12 states the scope. The appendices give the matrix proofs of rigidity (Appendix A), the comparison with the earlier package (Appendix B) and the verification record (Appendix C).

2 Definitions and conventions

Write an arbitrary two-qubit density matrix as

$$\rho = \frac{1}{4}\left(\mathbb{I}\otimes\mathbb{I} + \mathbf{a}\cdot\boldsymbol{\sigma}\otimes\mathbb{I} + \mathbb{I}\otimes\mathbf{b}\cdot\boldsymbol{\sigma} + \sum_{i,j=1}^{3} T_{ij}\sigma_i\otimes\sigma_j\right), \qquad \rho \succeq 0, \quad \operatorname{Tr}\rho = 1. \tag{1}$$

The correlation quantity in Eq. (3) of Ducuara–Susa–Skrzypczyk [1] is $F_3(\rho) = \|T\|_F$, with violation when $F_3 > 1$. Its identification with the optimized three-setting linear CJWR expression uses the measurement optimization of Costa–Angelo [6]. This is a particular steering inequality; $F_3 \le 1$ does not prove the existence of a local hidden-state model for all measurements. The orthonormal Bob triad is optimized here. If that triad is fixed and only Alice's three directions are optimized, the value is instead $\sum_j \|T\mathbf{e}_j\|/\sqrt{3}$, which can be smaller than $\|T\|_F$. Thus the matrix target is exactly the convention of Eq. (3), not an assertion about every fixed laboratory frame.

We use Alice's steering ellipsoid. Put $r = \|\mathbf{b}\|$. If $r < 1$, its centre is

$$\mathbf{c} = \frac{\mathbf{a} - T\mathbf{b}}{1 - r^2}, \qquad c = \|\mathbf{c}\|. \tag{2}$$

Whitening Bob produces the canonical state

$$\sigma = (\mathbb{I} \otimes (2\rho_B)^{-1/2})\rho(\mathbb{I} \otimes (2\rho_B)^{-1/2}), \qquad \sigma_B = \mathbb{I}/2. \tag{3}$$

Its Alice Bloch vector is $\mathbf{c}$; write its correlation matrix as S and its ellipsoid matrix as $Q = SS^{\mathsf{T}}$. These conventions are standard in the steering-ellipsoid construction [3, 4]. Swapping Alice and Bob gives an identical theorem for the other centre.

If $r = 1$, the state is a product $\rho_A \otimes |\beta\rangle\langle\beta|$. Its steering set is one point; we define its centre to be $\mathbf{a}$. This is a separate case, not a substitution into the zero denominator in (2).

The 2024 paper leaves the function bounding F_3 in terms of c undetermined and reports an apparent cutoff near 0.66 from random states. The earlier package [2] showed that the exact threshold is 2/3; Section 4.1 recalls an exact state with centre length above 0.66 and $F_3 > 1$.

3 A rank-safe geometric inequality

The following supplies an independent geometric route to the canonical bound used by the earlier proof. It is stronger than a bound involving c alone.

Theorem 1 (Directional matrix inequality)**.** *For every state* (1)*, without a rank restriction,*

$$M_A = (1 + \|\mathbf{a}\|^2 - \|\mathbf{b}\|^2 - \|T\|_F^2)\mathbb{I}_3 + 2(TT^{\mathsf{T}} - \mathbf{a}\mathbf{a}^{\mathsf{T}}) \succeq 0, \tag{4}$$

$$M_B = (1 + \|\mathbf{b}\|^2 - \|\mathbf{a}\|^2 - \|T\|_F^2)\mathbb{I}_3 + 2(T^{\mathsf{T}}T - \mathbf{b}\mathbf{b}^{\mathsf{T}}) \succeq 0. \tag{5}$$

For a unit vector $\mathbf{n}$*, equality* $\mathbf{n}^{\mathsf{T}}M_A\mathbf{n} = 0$ *holds exactly when* ρ *and the positive spin-flipped state defined below have orthogonal supports.*

Proof. Let $Y = \sigma_y$, $N = \mathbf{n} \cdot \boldsymbol{\sigma}$, and define

$$\widetilde{\rho} = (Y \otimes Y)\overline{\rho}(Y \otimes Y), \qquad \rho_{\mathbf{n}} = (N \otimes \mathbb{I})\widetilde{\rho}(N \otimes \mathbb{I}).$$

Complex conjugation preserves positivity. Both conjugating matrices are unitary for unit $\mathbf{n}$, so $\rho_{\mathbf{n}} \succeq 0$. The full spin flip sends $(\mathbf{a}, \mathbf{b}, T)$ to $(-\mathbf{a}, -\mathbf{b}, T)$, and conjugation by N acts on Alice's Bloch space by $R_{\mathbf{n}} = 2\mathbf{n}\mathbf{n}^{\mathsf{T}} - \mathbb{I}$. Pauli orthogonality gives the exact identity

$$4\,\mathrm{Tr}(\rho\rho_{\mathbf{n}}) = 1 + \|\mathbf{a}\|^2 - \|\mathbf{b}\|^2 - 2(\mathbf{a}\cdot\mathbf{n})^2 + 2\left\|T^{\mathsf{T}}\mathbf{n}\right\|^2 - \|T\|_F^2 = \mathbf{n}^{\mathsf{T}}M_A\mathbf{n}. \tag{6}$$

The left side is nonnegative because it is the trace of $\rho^{1/2}\rho_{\mathbf{n}}\rho^{1/2} \succeq 0$. Homogeneity extends nonnegativity from unit vectors to all real vectors. Exchange the two qubits for M_B.

If the overlap is zero, the positive matrix $\rho^{1/2}\rho_{\mathbf{n}}\rho^{1/2}$ is zero. Equivalently, $\rho_{\mathbf{n}}^{1/2}\rho^{1/2} = 0$, giving orthogonal supports and $\rho\rho_{\mathbf{n}} = 0$. The converse is immediate. No inverse is used. □

Theorem 2 (Sharp centre and radial-width envelope)**.** *For a canonical state with centre* $\mathbf{c}$*, choose* $\mathbf{n} = \mathbf{c}/c$ *when* $c > 0$*; when* $c = 0$*, choose any unit direction. Put*

$$t = \left\|S^{\mathsf{T}}\mathbf{n}\right\|, \qquad q = t^2 = \mathbf{n}^{\mathsf{T}}Q\mathbf{n}.$$

Then

$$0 \le t \le 1 - c, \qquad \mathrm{tr}\,Q \le 1 - c^2 + 2t^2. \tag{7}$$

For every $0 \le c \le 1$ *and* $0 \le t \le 1 - c$*, the upper bound is attained. Consequently, if* $\delta = (1-c)(3-c) - \mathrm{tr}\,Q$*, then*

$$\delta \ge 2\big((1-c)^2 - q\big) \ge 0. \tag{8}$$

Here t is the support half-width of the ellipsoid in the centre direction. For a tilted ellipsoid it need not be a semiaxis or its radius along that ray.

Proof. Specialize (4) to $\mathbf{b} = 0$, $\mathbf{a} = \mathbf{c}$, and evaluate it on $\mathbf{n}$. This gives the second inequality in (7). Every conditional Bloch vector $\mathbf{c} + S\mathbf{u}$, $\|\mathbf{u}\| \le 1$, has norm at most one. Maximizing its projection on $\mathbf{n}$ gives $c + \left\|S^{\mathsf{T}}\mathbf{n}\right\| = c + t \le 1$. This argument is valid also at $t = 0$. Substitute $t \le 1 - c$ to obtain (8).

For sharpness, define

$$a_\pm = \frac{1 + c \pm t}{4}, \qquad d_\pm = \frac{1 - c \pm t}{4}, \qquad w = \sqrt{a_+ d_+}, \quad v = \sqrt{a_- d_-},$$

and, in the order $00, 01, 10, 11$, set

$$\sigma_{c,t} = \begin{pmatrix} a_+ & 0 & 0 & w \\ 0 & a_- & v & 0 \\ 0 & v & d_- & 0 \\ w & 0 & 0 & d_+ \end{pmatrix}. \tag{9}$$

All four diagonal parameters are nonnegative. After grouping the pairs $00, 11$ and $01, 10$, the matrix is the direct sum of two positive matrices of rank at most one, including a zero block at an endpoint. Its eigenvalues are $(1 + t)/2, (1 - t)/2, 0, 0$. Its marginals are $(\mathbb{I} + cZ)/2$ and $\mathbb{I}/2$, and

$$S = \operatorname{diag}(2(w + v), 2(v - w), t).$$

Hence $\left\|S^{\mathsf{T}}\mathbf{e}_z\right\|^2 = t^2$ and $\operatorname{tr} Q = 8(w^2 + v^2) + t^2 = 1 - c^2 + 2t^2$. The construction includes the zero-block endpoints, the rank-one Bell state $(c, t) = (0, 1)$, and the product $|0\rangle\langle 0| \otimes \mathbb{I}/2$ at $(1, 0)$. □

4 The exact global correlation envelope

The envelope, the joint upper bound and the attaining family in this section are results of the earlier package [2]. We recall them with their proofs, because the new results build on the same identities.

Lemma 3 (Exact filter identity and marginal bound)**.** *Let $\sigma_B = \mathbb{I}/2$, and let $F \succeq 0$ satisfy $\operatorname{Tr} F^2 = 2$. Write $\rho = (\mathbb{I} \otimes F)\sigma(\mathbb{I} \otimes F)$ and $r = \|\mathbf{b}\|$. Then*

$$\|T\|_F^2 = (1 - r^2)(\|S\|_F^2 - c^2) + \|\mathbf{a}\|^2, \tag{10}$$

$$\|\mathbf{a}\| \le c + (1 - c)r. \tag{11}$$

These identities and inequalities remain valid for singular F when c means the prefilter canonical centre.

Proof. For any complex 4×4 matrix X, define $E(X) = \operatorname{Tr}(X^2) - \operatorname{Tr}((\operatorname{Tr}_B X)^2)$. For every 2×2 matrix F,

$$E((\mathbb{I} \otimes F)X(\mathbb{I} \otimes F)) = \det(F)^2 E(X). \tag{12}$$

To verify this, write X in Bob-valued blocks X_{ij} and use $E(X) = \sum_{ij}[\operatorname{Tr}(X_{ij}X_{ji}) - \operatorname{Tr} X_{ij} \operatorname{Tr} X_{ji}]$. Polarizing the 2×2 determinant identity under $A \mapsto FAF$ proves the equality for each pair of blocks. It is a polynomial identity and requires neither an inverse nor positivity.

Here $\rho_B = F^2/2$, so $\det(F)^2 = 1 - r^2$. The Pauli purity identity $4 \operatorname{Tr} \rho^2 = 1 + \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 + \|T\|_F^2$ turns (12) into (10).

The smallest eigenvalue of F^2 is $1-r$, so $K = F^2 - (1-r)\mathbb{I} \succeq 0$ and $\operatorname{Tr} K = 2r$. Put $R = \operatorname{Tr}_B[\sigma(\mathbb{I}\otimes K)]$. Partial-trace cyclicity writes R as the partial trace of a positive congruence, proving $R \succeq 0$. Moreover $\operatorname{Tr} R = r$ and $\rho_A = (1-r)\sigma_A + R$. The Bloch-vector length of a positive qubit matrix is bounded by its trace. The triangle inequality therefore gives (11), including $r = 0, 1$.

For the centre identification, let $B = \mathbf{b}\cdot\boldsymbol{\sigma} = F^2 - \mathbb{I}$. The filter commutes with B, and $B^2 = r^2\mathbb{I}$, so $FBF = B + r^2\mathbb{I}$. Taking Pauli expectations and using partial-trace cyclicity gives

$$\mathbf{a} = \mathbf{c} + S\mathbf{b}, \qquad T\mathbf{b} = S\mathbf{b} + r^2\mathbf{c}, \qquad \mathbf{a} - T\mathbf{b} = (1-r^2)\mathbf{c}.$$

For $r < 1$ this proves that the whitening centre is exactly (2). It requires no inverse of the joint state. □

Theorem 4 (Joint bound and global envelope)**.** *For an actual output state with $r < 1$ and centre length c,*

$$F_3(\rho)^2 \le H_c(r) := (1-r^2)(3-4c) + [c + (1-c)r]^2. \tag{13}$$

For every pair $0 \le c \le 1$, $0 \le r < 1$, equality is attained. At $r = 1$ the sharp bound at the actual output centre is instead c^2.

Over all states with specified actual centre length c,

$$\sup F_3^2 = G(c) = \begin{cases} 1 + \dfrac{(2-3c)^2}{2-2c-c^2}, & 0 \le c < 2/3, \\ 1, & 2/3 \le c \le 1. \end{cases} \tag{14}$$

The supremum is attained for $c < 2/3$ and $c = 1$, and is not attained for $2/3 \le c < 1$. There exists an F_3-violating state at centre length c if and only if $c < 2/3$.

Proof. For $r < 1$, use (3) and apply Theorem 2 and Lemma 3. They give $\|S\|_F^2 - c^2 \le 3 - 4c$ and (13).

Put $d = 1-c$ and $D(c) = 2-2c-c^2$. Then $H_c(r) = 3 - 4c + c^2 + 2cd\,r - D(c)r^2$. On $0 \le c \le 2/3$, $D(c) > 0$ and

$$G(c) - H_c(r) = D(c)\left(r - \frac{c(1-c)}{D(c)}\right)^2. \tag{15}$$

The optimizer $r_* = c(1-c)/D(c)$ belongs to $[0,1)$ exactly when $c < 2/3$. At $c = 2/3$ it equals one. For $2/3 \le c \le 1$ the exact identity

$$1 - H_c(r) = (1-r)^2(4c - 2 - c^2) + r(1-r)(6c-4) \tag{16}$$

is nonnegative. Its first coefficient is strictly positive throughout that interval; thus $H_c(r) < 1$ if $r < 1$.

If $r = 1$, positivity and the pure Bob marginal force the product form: indeed the compression of ρ to Bob's orthogonal marginal support has trace zero, hence is zero, and positivity annihilates its off-support rows and columns. Then $T = \mathbf{a}\mathbf{b}^{\mathsf{T}}$ and $F_3^2 = \|\mathbf{a}\|^2 = c^2$. No extra maximizer appears when $c < 1$.

The explicit family in the next subsection attains (13) for every $r < 1$. Choosing $r = r_*$ proves attainment on the first branch. Letting $r \uparrow 1$ proves sharpness of the flat branch. At $c = 1$, a pure product state attains one. Finally $G(c) > 1$ precisely when $c < 2/3$. □

4.1 Explicit sharp states and exact counterexamples

The following family is a Bob-filtered maximally obese canonical state:

$$\rho_{c,r} = \frac{1}{2}\begin{pmatrix} 1+r & 0 & 0 & \sqrt{(1-c)(1-r^2)} \\ 0 & c(1-r) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{(1-c)(1-r^2)} & 0 & 0 & (1-c)(1-r) \end{pmatrix}, \quad 0 \le r < 1. \tag{17}$$

It is positive semidefinite and trace one. Its Bloch data are

$$\mathbf{a} = [c + (1-c)r]\mathbf{e}_z, \quad \mathbf{b} = r\mathbf{e}_z, \quad T = \operatorname{diag}\left(\sqrt{(1-c)(1-r^2)}, -\sqrt{(1-c)(1-r^2)}, 1-c+cr\right).$$

Substitution into (2) gives $\mathbf{c} = c\mathbf{e}_z$ and $F_3^2 = H_c(r)$. The canonical representative is $\sigma_{c,1-c}$.

At $c = 3/5$ the best filter is $r = 6/11$, giving

$$\rho = \frac{1}{22}\begin{pmatrix} 17 & 0 & 0 & \sqrt{34} \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{34} & 0 & 0 & 2 \end{pmatrix}, \qquad F_3^2 = \frac{12}{11} > 1.$$

Its canonical representative has $F_3^2 = 24/25 < 1$. This explicitly excludes the tempting but invalid argument that canonical filtering preserves F_3.

At $c = 53/80 = 0.6625 > 0.66$, the exact optimizer is $r = 1431/1511$ and $F_3^2 = 1512/1511 > 1$. Thus the reported decimal 0.66 can only be interpreted as numerical evidence near the true cutoff.

For $2/3 \le c < 1$, the states (17) approach $|00\rangle\langle 00|$ as $r \uparrow 1$. That limit has centre length one, not c. The fixed-c set is not closed at this singular marginal boundary; this explains why compactness of the entire density-matrix set does not turn the supremum into a maximum.

5 The full correlation range at fixed centre and marginal

The upper envelope has a complementary sharp lower bound. Together they determine every possible value of the correlation norm, not merely its largest value. This also identifies a region in which a nonzero centre forces a strictly positive measured correlation.

Theorem 5 (Complete joint range)**.** *Fix $0 \le c \le 1$ and $0 \le r < 1$. Among all two-qubit density matrices with Alice steering-centre length c and Bob Bloch length r, the set of attainable squared correlations is exactly*

$$\left[\left(\max\{0, c(1+r)-1\}\right)^2,\ (1-r^2)(3-4c) + \left(c+(1-c)r\right)^2\right]. \tag{18}$$

Both endpoints are attained. The lower endpoint is attained by a state diagonal in a product basis. Every lower-endpoint state is locally unitarily equivalent to the displayed diagonal witness below. At $r = 1$, the entire range is instead the singleton $\{c^2\}$.

Proof. Choose unit vectors $\mathbf{n}, \mathbf{m}$ so that the centre is $c\mathbf{n}$ and $\mathbf{b} = r\mathbf{m}$. When a vector vanishes, choose its direction arbitrarily; the argument does not divide by c or r. Put $q = \mathbf{n}^{\mathsf{T}} T \mathbf{m}$. The probability of obtaining both minus outcomes in the corresponding local spin measurements is nonnegative:

$$1 - \mathbf{a}\cdot\mathbf{n} - r + q \ge 0.$$

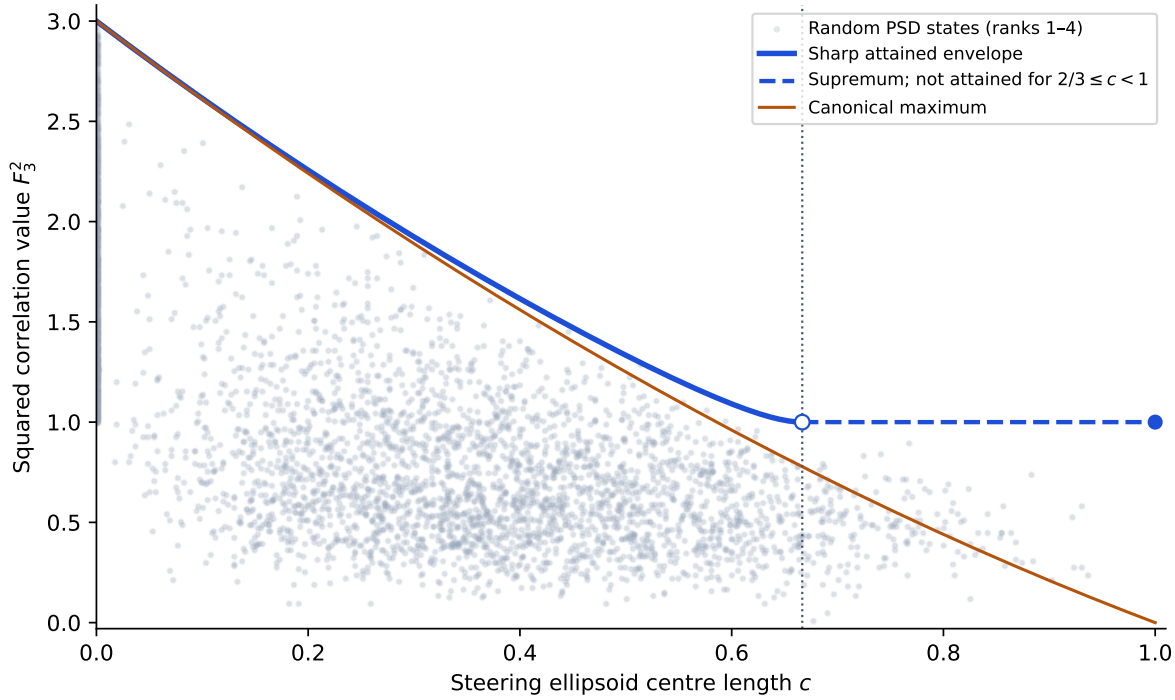


Figure 1: The exact envelope, with random states shown only as numerical diagnostics. The dashed branch is not attained before $c = 1$. Random sampling is especially insensitive to the narrow violation region near $2/3$.

The centre relation gives $\mathbf{a} \cdot \mathbf{n} = (1 - r^2)c + rq$, hence

$$0 \le (1 - r)\big(q - [c(1 + r) - 1]\big). \tag{19}$$

Since $r < 1$, $q \ge c(1 + r) - 1$. Cauchy–Schwarz gives $|q| \le \|T\|_F$, proving the lower bound. The upper bound is the sharp joint envelope proved earlier.

For attainment, put

$$x = \max\{0, c(1 + r) - 1\}, \qquad a_0 = c(1 - r^2) + rx.$$

Take $\mathbf{a} = a_0\mathbf{e}_z$, $\mathbf{b} = r\mathbf{e}_z$, and $T = \mathrm{diag}(0, 0, x)$. The associated density matrix is

$$\rho_{c,r}^{-} = \frac{1}{4}\,\mathrm{diag}(1 + a_0 + r + x,\ 1 + a_0 - r - x,\ 1 - a_0 + r - x,\ 1 - a_0 - r + x). \tag{20}$$

If $x = 0$, then $0 \le a_0 \le 1 - r$, so all entries are nonnegative. If $x > 0$, then $0 < x \le r$, $a_0 = 1 - r + x$, and its eigenvalues are

$$\frac{1 + x}{2}, \quad \frac{1 - r}{2}, \quad \frac{r - x}{2}, \quad 0.$$

Its centre is exactly $c\mathbf{e}_z$ and $F_3^2 = x^2$. These checks include all vanishing diagonal entries, without assuming a strictly positive state.

Let $\rho_{c,r}^{+}$ be the upper witness in Eq. (17), in the same coordinate convention. Both witnesses have the same $\mathbf{b} = r\mathbf{e}_z$ and the same vector $\mathbf{a} - T\mathbf{b} = (1 - r^2)c\mathbf{e}_z$. Every convex combination

$$\rho_\alpha = (1 - \alpha)\rho_{c,r}^{-} + \alpha\rho_{c,r}^{+}, \qquad 0 \le \alpha \le 1,$$

therefore has the same centre and Bob marginal. Write $d = 1 - c$ and $z = d + cr$. Direct computation gives

$$F_3(\rho_\alpha)^2 = 2\alpha^2 d(1 - r^2) + \big[x + \alpha(z - x)\big]^2. \tag{21}$$

Here $x \geq 0$ and $z - x \geq 0$; for $x > 0$, the latter difference is $2(1-c)$. Thus the expression is continuous and nondecreasing from the lower endpoint to the upper endpoint. It supplies every intermediate value.

For the equality classification, if $x > 0$ then equality in the lower bound forces $q = x$ and equality in the Frobenius Cauchy–Schwarz inequality. Consequently $T = x\mathbf{n}\mathbf{m}^{\mathsf{T}}$, and the centre relation fixes $\mathbf{a} = a_0\mathbf{n}$. If $x = 0$, equality means $T = 0$, and the centre relation again fixes $\mathbf{a} = c(1-r^2)\mathbf{n}$. Independent local changes of spin axes give precisely Eq. (20). Finally, when $r = 1$, positivity forces $\rho = \rho_A \otimes |\beta\rangle\langle\beta|$; its point centre has length c, and its squared correlation is c^2. □

The zero-correlation region is exactly $c(1+r) \leq 1$. A state with $T = 0$ need not be a product: when both local vectors are nonzero, the product state would have $T = \mathbf{a}\mathbf{b}^{\mathsf{T}}$. The lower witness is nevertheless separable because it is diagonal in a product basis.

For example, at $c = 3/4$ and $r = 1/2$, the lower witness is

$$\rho^{-}_{3/4,1/2} = \mathrm{diag}(9/16, 1/4, 3/16, 0), \qquad F_3^2 = 1/64,$$

while the upper value is $49/64$. All values between them occur at exactly the same centre length and Bob marginal. At $c = 1$, the two endpoints coincide at r^2; the allowed states have pure Alice marginal and are products. At $r = 0$, the range reduces to $[0, (1-c)(3-c)]$.

Corollary 6 (A quantitative directional consequence)**.** *For every state with $r < 1$, centre $c\mathbf{n}$, and Bob vector $r\mathbf{m}$,*

$$\mathbf{n}^{\mathsf{T}} T \mathbf{m} \geq c(1+r) - 1, \qquad c(1+r) \leq 1 + \|T\|_F .$$

If $x = c(1+r) - 1 > 0$ and $F_3^2 - x^2 \leq \eta$, then

$$\left\| T - x\mathbf{n}\mathbf{m}^{\mathsf{T}} \right\|_F^2 \leq \eta, \qquad \|\mathbf{a} - a_0\mathbf{n}\|^2 \leq r^2\eta. \tag{22}$$

Proof. The first assertions were proved above. Since $q \geq x > 0$,

$$\left\| T - x\mathbf{n}\mathbf{m}^{\mathsf{T}} \right\|_F^2 = \|T\|_F^2 - 2xq + x^2 \leq \|T\|_F^2 - x^2 \leq \eta.$$

Subtracting the common centre equations gives $\mathbf{a} - a_0\mathbf{n} = r(T - x\mathbf{n}\mathbf{m}^{\mathsf{T}})\mathbf{m}$; the Frobenius bound proves the second assertion. □

This yields an explicit state-distance consequence in the same axes. For ρ^{-} with the same centre and Bob vector,

$$\|\rho - \rho^{-}\|_2^2 = \tfrac{1}{4}\left(\|\mathbf{a} - a_0\mathbf{n}\|^2 + \left\| T - x\mathbf{n}\mathbf{m}^{\mathsf{T}} \right\|_F^2\right) \leq \frac{1+r^2}{4}\eta, \qquad \|\rho - \rho^{-}\|_1^2 \leq (1+r^2)\eta.$$

The norm $\|\cdot\|_2$ is the Hilbert–Schmidt norm. The same conclusion holds in the $x = 0$ region with $\eta = F_3^2$, since then the correlation matrix of the witness vanishes. Thus lower-endpoint rigidity has a direct square-root modulus, including the interface $c(1+r) = 1$.

6 Complete canonical correlation intervals and a sharp success-probability frontier

The centre and radial-width bound determines an entire attainable interval, not only its upper endpoint. A separate operational refinement keeps track of the probability of the filtering event. Throughout this section a canonical input has Bloch data $(c\mathbf{n}, 0, S)$, where $\|\mathbf{n}\| = 1$ and $0 \leq c \leq 1$.

When $c = 0$, the direction $\mathbf{n}$ is a specified reference direction; it is not claimed to be determined by the vanishing centre. Set

$$t = \left\| S^{\mathsf{T}}\mathbf{n} \right\|, \qquad \tau = \|S\|_F^2 .$$

All Kraus probabilities below refer to the canonical *input* state.

Theorem 7 (Complete canonical correlation interval)**.** *The pair (c, t) is feasible exactly when $0 \le t \le 1 - c$. At every feasible pair, the complete set of squared correlation norms is*

$$\boxed{t^2 \le \tau \le 1 - c^2 + 2t^2,} \tag{23}$$

with every value in this closed interval attained by a density matrix. Both endpoints and all singular parameter cases are included.

The lower endpoint is attained exactly when $S = \mathbf{n}\mathbf{u}^{\mathsf{T}}$ for some $\|\mathbf{u}\| = t$. Thus the lower-endpoint state is unique up to local unitaries. The upper-endpoint state is also unique up to local unitaries and has the matrix form displayed below. Its spectrum is $\{(1 + t)/2, (1 - t)/2, 0, 0\}$.

Proof. A local rotation puts $\mathbf{n} = \mathbf{e}_z$; a Bob rotation puts $S^{\mathsf{T}}\mathbf{n} = t\mathbf{e}_z$. Positivity of the conditional qubit with Bob direction $\mathbf{e}_z$ gives $c + t \le 1$. The lower bound follows from

$$\tau - t^2 = \left\| (\mathbb{I} - \mathbf{n}\mathbf{n}^{\mathsf{T}})S \right\|_F^2 \ge 0,$$

and equality is precisely the stated rank-one condition. Its diagonal representative is physical for exactly $t \le 1 - c$. For the upper bound, let

$$U = (\mathbf{n} \cdot \boldsymbol{\sigma})\sigma_y \otimes \sigma_y, \qquad \sigma_{\mathbf{n}} = U\overline{\sigma}U^{\dagger}.$$

Both σ and $\sigma_{\mathbf{n}}$ are positive semidefinite, so their trace product is nonnegative. Pauli orthogonality gives

$$4\,\mathrm{Tr}(\sigma\sigma_{\mathbf{n}}) = 1 - c^2 + 2t^2 - \tau \ge 0.$$

This argument does not require either state to be invertible.

For explicit attainability define

$$a = \frac{1 + c + t}{4}, \quad b = \frac{1 + c - t}{4}, \quad e = \frac{1 - c - t}{4}, \quad d = \frac{1 - c + t}{4}.$$

All four numbers are nonnegative. For $0 \le \alpha \le 1$ set

$$\sigma_{c,t,\alpha} = \begin{pmatrix} a & 0 & 0 & \alpha\sqrt{ad} \\ 0 & b & \alpha\sqrt{be} & 0 \\ 0 & \alpha\sqrt{be} & e & 0 \\ \alpha\sqrt{ad} & 0 & 0 & d \end{pmatrix}. \tag{24}$$

Its two 2×2 blocks have determinants $(1 - \alpha^2)ad$ and $(1 - \alpha^2)be$. Hence it is a density matrix, including when a diagonal entry vanishes. Its Bloch data are

$$\mathbf{c} = c\mathbf{e}_z, \quad \mathbf{b} = 0, \quad S = \mathrm{diag}(2\alpha(\sqrt{ad} + \sqrt{be}), 2\alpha(\sqrt{be} - \sqrt{ad}), t),$$

so

$$\tau = t^2 + \alpha^2(1 - c^2 + t^2). \tag{25}$$

For $1 - c^2 + t^2 > 0$, any requested τ in (23) is obtained by choosing $\alpha = \sqrt{(\tau - t^2)/(1 - c^2 + t^2)}$. If that denominator vanishes, $c = 1, t = 0$ and the interval consists of the single value zero.

For completeness, consider upper-endpoint rigidity. In the chosen axes, every canonical state has block form

$$\sigma = \begin{pmatrix} A & C \\ C^\dagger & D \end{pmatrix}, \qquad A = \mathrm{diag}(a,b), \quad D = \mathrm{diag}(e,d), \quad \mathrm{Tr}\, C = 0.$$

If $0 < t < 1 - c$, both diagonal blocks are positive definite. Block positivity is equivalent to $C = A^{1/2} K D^{1/2}$ with K a contraction. The matrix $(|K_{ij}|^2)$ has row and column sums at most one. The elementary rearrangement inequality for such a 2×2 matrix gives

$$\|C\|_F^2 \le ad + be.$$

One way to verify the equality condition is to extend the matrix of squared moduli to a doubly stochastic matrix. Because all four weights are strictly positive, equality permits no added entry. Every doubly stochastic 2×2 matrix is a convex combination of the two permutation matrices; the difference between their weighted sums is $(a-b)(d-e) = t^2/4 > 0$. Equality therefore forces K to be antidiagonal with both nonzero entries of modulus one. Alice and Bob rotations about z remove its two phases, giving (24) with $\alpha = 1$.

If $t = 0$ and $c < 1$, the two blocks are scalar matrices. Equality gives $C = \sqrt{ad}\, V$, where V is a traceless unitary. Every traceless 2×2 unitary is a scalar phase times a traceless Hermitian unitary; Bob conjugation and an Alice phase rotation reduce it to σ_x. This again gives the displayed upper-endpoint representative. If $t = 1 - c > 0$, the diagonal entry e vanishes. Positivity forces its entire row and column to vanish; the remaining coherence has modulus at most $\sqrt{ad}$, and upper equality fixes that modulus. The cases $(c,t) = (0,1)$ and $(1,0)$ follow directly from the same zero-row rule. Thus no inverse of a zero diagonal block was used. At $\alpha = 1$, the two blocks have traces $(1+t)/2$ and $(1-t)/2$ and determinant zero, proving the eigenvalue claim. □

Ranks in the explicit interpolation. For $c < 1$ and $t < 1 - c$, the state (24) has rank four for $\alpha < 1$ and rank two for $\alpha = 1$. For $0 < c < 1$ and $t = 1 - c$, these ranks are three and two. At $(c,t) = (0,1)$ they are two and one. At $(c,t) = (1,0)$ the matrix is the fixed rank-two product state $|0\rangle\langle 0| \otimes \mathbb{I}/2$, for every α.

Lemma 8 (Exact filtering probability)**.** *Let $\sigma_B = \mathbb{I}/2$ and let G be a nonzero-probability Bob Kraus operator, $G^\dagger G \preceq \mathbb{I}$. Its success probability is*

$$p = \mathrm{Tr}[(\mathbb{I} \otimes G)\sigma(\mathbb{I} \otimes G^\dagger)] = \tfrac{1}{2}\,\mathrm{Tr}(G^\dagger G) > 0.$$

Writing $F = G/\sqrt{p}$ gives the unique normalized effect representation

$$F^\dagger F = \mathbb{I} + \mathbf{x} \cdot \boldsymbol{\sigma}, \qquad r = \|\mathbf{x}\| \le 1.$$

For a fixed normalized effect, the exact maximal success probability is

$$p_{\max}(r) = \frac{1}{1+r}. \tag{26}$$

Every probability $0 < p \le p_{\max}(r)$ is attained by $G = \sqrt{p}\, F$. Its normalized output is exactly $(\mathbb{I} \otimes F)\sigma(\mathbb{I} \otimes F^\dagger)$.

Proof. The trace identity follows by tracing out Alice and using $\sigma_B = \mathbb{I}/2$. The positive matrix $F^\dagger F$ has trace two and therefore has the indicated Bloch representation, with eigenvalues $1 \pm r$. The Kraus condition is

$$\mathbb{I} - pF^\dagger F \succeq 0 \quad \Longleftrightarrow \quad p(1+r) \le 1.$$

This proves both necessity and attainability, including $r = 1$. The scalar factor p cancels from the normalized output. If $p > 1/2$, the inequality forces $r < 1$, so G is invertible. Conversely a singular nonzero qubit Kraus operator has $r = 1$ and success at most $1/2$. □

For a required success probability $p_0 \in [1/2, 1]$, define

$$R(p_0) = \frac{1-p_0}{p_0}, \qquad D(c) = 2 - 2c - c^2, \tag{27}$$

$$H_c(r) = (1-r^2)(3-4c) + [c + (1-c)r]^2, \tag{28}$$

$$r_*(c) = \frac{c(1-c)}{D(c)} \quad (0 \le c \le 2/3), \qquad \widehat{r}(c, p_0) = \begin{cases} \min\{R(p_0), r_*(c)\}, & c \le 2/3, \\ R(p_0), & c > 2/3. \end{cases} \tag{29}$$

The definition of r_* at $c = 2/3$ is regular and gives $r_* = 1$.

Theorem 9 (Sharp probability-constrained correlation envelope)**.** *Over all canonical input density matrices with centre length $c \in [0, 1]$ and all Bob Kraus operators of success probability at least $p_0 \in [1/2, 1]$, the exact greatest squared correlation norm is*

$$\boxed{\mathcal{G}(c, p_0) = H_c(\widehat{r}(c, p_0)).} \tag{30}$$

This is a maximum, including all endpoints. An explicit maximizer is

$$\sigma_c = \frac{c}{2}|00\rangle\langle 00| + \frac{1}{2}(|01\rangle + \sqrt{1-c}|10\rangle)(\langle 01| + \sqrt{1-c}\langle 10|), \tag{31}$$

$$G_{c,p_0} = \sqrt{p_0}\ \mathrm{diag}(\sqrt{1-\widehat{r}}, \sqrt{1+\widehat{r}}). \tag{32}$$

The actual success probability of this witness is exactly p_0.

Proof. Normalize any admissible Kraus operation as in Lemma 8. If its actual success is $p \ge p_0$, then $p_0(1+r) \le p(1+r) \le 1$, hence $0 \le r \le R(p_0)$. The exact correlation identity and positivity give

$$F_3^2 = (1-r^2)(\tau - c^2) + \|\mathbf{c} + S\mathbf{x}\|^2 \le (1-r^2)(3-4c) + [c + (1-c)r]^2 = H_c(r).$$

The canonical bound used here is $\tau \le 3 - 4c + c^2$. The marginal bound follows by writing $\mathbf{c} + S\mathbf{x} = (1-r)\mathbf{c} + r(\mathbf{c} + S\mathbf{u})$, with $\mathbf{u} = \mathbf{x}/r$ when $r > 0$; the conditional endpoint $\mathbf{c} + S\mathbf{u}$ lies in the Bloch ball. The case $r = 0$ follows directly. The identity and inequalities hold also for $r = 1$ by direct substitution.

Expanding gives

$$H_c(r) = 3 - 4c + c^2 + 2c(1-c)r - D(c)r^2.$$

For $c \le 2/3$, $D(c) > 0$ and the quadratic reaches its maximum at $r_*(c)$; clipping to $[0, R]$ gives (29). For $c \ge 2/3$, the quadratic is nondecreasing on $[0, 1]$. Indeed if $D \ge 0$, its derivative is bounded below by $2[c(1-c) - D] = 2(3c - 2) \ge 0$, and if $D < 0$ its derivative is nonnegative directly. The endpoint R is therefore optimal.

The state (31) is a positive sum of two rank-one operators, has trace one, has Bob marginal $\mathbb{I}/2$, and has centre length c. Direct multiplication by the displayed diagonal filter gives $F_3^2 = H_c(\widehat{r})$. Its normalized effect has radius $\widehat{r}$; $p_0(1+\widehat{r}) \le 1$ proves the actual Kraus condition. Its success is p_0 by the canonical trace identity. This proves the maximum over the full physical class, not only over a scalar relaxation. □

Corollary 10 (Exact activation frontier)**.** *There exists a canonical input of centre length c and a Bob Kraus operation of success probability at least $p_0 \in [1/2, 1]$ whose output has $F_3 > 1$ if and only if*

$$\boxed{c < C_{\mathrm{crit}}(p_0) := \frac{2}{1 + p_0 + \sqrt{(1-p_0)^2 + 2}}.} \tag{33}$$

The equality curve is nonviolating. It interpolates between $C_{\mathrm{crit}}(1/2) = 2/3$ and $C_{\mathrm{crit}}(1) = 2 - \sqrt{2}$.

Proof. Let $E(c) = 2 - 4c + c^2$. The exact factorization is

$$H_c(r) - 1 = (1 - r)[D(c)r + E(c)]. \tag{34}$$

If there is a violating point with $r \leq R$, then $c < 2/3$ by the global nonviolation bound, so $D(c) > 0$. The linear second factor must be positive and is increasing in r. It follows that

$$g(c, p_0) := p_0[D(c)R(p_0) + E(c)] = 2 - 2(1 + p_0)c + (2p_0 - 1)c^2 > 0.$$

For $p_0 > 1/2$, the converse follows by evaluating the endpoint:

$$H_c(R) - 1 = \frac{(2p_0 - 1)g(c, p_0)}{p_0^2}.$$

At $p_0 = 1/2$, this endpoint identity is identically zero and cannot establish activation. Instead $g = 2 - 3c > 0$ means $c < 2/3$; the interior vertex $r_*(c) < 1 = R$ gives $H_c(r_*) = G(c) > 1$.

Put $s = \sqrt{(1 - p_0)^2 + 2}$. Then

$$2g(c, p_0) = [2 - (1 + p_0 + s)c][2 - (1 + p_0 - s)c].$$

For $0 \leq c \leq 1$ and $1/2 \leq p_0 \leq 1$, the second factor is strictly positive: $s \geq 1$, hence $(1+p_0-s)c \leq 1$. The sign of g is therefore exactly the sign of the first factor, proving the claimed strict threshold. □

Nonbinding budgets and equality for $p_0 > 1/2$. For $0 \leq c < 2/3$, the probability constraint leaves the unrestricted maximum unchanged exactly when

$$p_0 \leq p_*(c) := \frac{1}{1 + r_*(c)} = \frac{D(c)}{2 - c - 2c^2}. \tag{35}$$

Above that value the maximum is the endpoint expression $H_c(R(p_0))$. For $2 - \sqrt{2} < c < 2/3$, the largest allowed success floor for strict activation is equivalently

$$p_0 < \frac{D(c)}{2c(1 - c)}.$$

At $c = 2 - \sqrt{2}$, activation occurs for every $p_0 < 1$ and fails at $p_0 = 1$. For $c < 2 - \sqrt{2}$, the displayed canonical input (31) itself already violates.

For $0 < c < 1$ and $p_0 > 1/2$, an extremizer must saturate the sharp joint centre/radius bound at its uniquely optimal radius $\widehat{r} < 1$. The canonical input is therefore a local-unitary copy of (31). When $\widehat{r} > 0$, its normalized effect points toward the pure contact of that canonical ellipsoid; in the displayed axes it is $\mathbb{I} - \widehat{r}\sigma_z$. A post-filter Bob unitary is unrestricted. When $\widehat{r} = 0$, the normalized effect is the identity. In a strictly binding case $\widehat{r} = R$, the actual success of every maximizer is exactly p_0. In a nonbinding case $\widehat{r} = r_*$, the actual success can be any $q \in [p_0, p_*(c)]$, by scalar rescaling of the same normalized filter. At the crossover these two descriptions agree. For $c = 0$, the canonical maximizers are maximally entangled pure states and the optimal normalized effect is the identity. For $c = 1$, the canonical state is $a \otimes \mathbb{I}/2$ with a pure, and $F_3^2 = r^2$; any filter direction is allowed, with $r = R$.

Singular endpoints and the meaning of the centre constraint. At $p_0 = 1/2$, all radii through $r = 1$ are feasible. For $c < 2/3$, the optimum still uses the invertible radius $r_* < 1$. For $2/3 \leq c \leq 1$, the maximum is one and is attained at $r = 1$ by a singular Kraus operator selecting a pure contact point. More generally, among canonical inputs in this range, a singular filter attains one precisely when its selected conditional Alice state is pure; the normalized output is then a

pure product. Many canonical input orbits with pure ellipsoid contacts can therefore attain this singular maximum. No unique obese-input classification is claimed there.

For $p_0 > 1/2$, all admissible filters are invertible and preserve the steering ellipsoid and its centre. At $p_0 = 1/2$, the singular maximizing output has centre length one, even if the specified *input* centre is $c < 1$. This does not contradict the nonattainment of the global fixed-*output* centre envelope for $2/3 \le c < 1$. If the success floor is instead $0 < p_0 < 1/2$, every normalized radius in $[0, 1]$ is feasible; the maximum is $G(c)$, interpreted as an attained value one on the large-centre branch, and the activation threshold remains $c < 2/3$.

Exact formalization boundary for this section. The complete physical interval, including its interpolation witness and lower-endpoint equality condition, is checked in `MilneGeometryInterval.lean`, principally by `geometryPhysicalSkewSpectrum_eq_interval` and `geometry_radial_lower_equality_iff`. The upper-endpoint local-unitary classification and the listed interpolation ranks are written proofs in this section.

`MilneKrausProbability.lean` checks actual complex Kraus positivity, normalization, success probability, and output density. Its principal lemmas are `scaledKraus_admissible_iff`, `fixed_filter_maximal_success`, `canonicalKrausProbability_is_actual`, `normalizedKrausOutput_density`, and `canonical_complex_joint_bound`. `MilneProbabilityScalar.lean` checks the clipped optimizer and exact activation curve, including the separate $p_0 = 1/2$ argument, through `successOptimalRadius_maximizes` and `successEnvelope_violation_iff_center`. `MilneProbabilityEnvelope.lean` states the optimization over *actual* density matrices and contracting complex Kraus operators. Its main theorems are `successPhysicalSpectrum_isGreatest` and `successPhysicalSpectrum_violation_iff`. The exact saturation below probability one half is also checked by `successPhysicalSpectrum_low_probability` and `looseSuccessPhysicalSpectrum_isGreatest`; singularity exclusion above one half is checked by `KrausAdmissible_det_ne_zero_of_success_gt_half`. The additional local-unitary equality classification and probability nonbinding descriptions are written consequences; they are not represented as separately checked Lean theorems here.

6.1 An exact physical filtering experiment with rational matrices

The probability frontier can be illustrated without numerical diagonalization. Take the canonical rank-two input

$$\sigma = \frac{1}{50}\begin{pmatrix} 25 & 0 & 0 & 15 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 15 & 0 & 0 & 9 \end{pmatrix}, \qquad c = \frac{16}{25}, \qquad S = \operatorname{diag}\left(\frac{3}{5}, -\frac{3}{5}, \frac{9}{25}\right).$$

Its squared correlation is $531/625 < 1$. The contracting Kraus operator

$$G = \operatorname{diag}(1, 5/13), \qquad G^\dagger G \preceq I,$$

has success probability $p = 97/169$. Conditional on success, its output is

$$\rho = \frac{1}{194}\begin{pmatrix} 169 & 0 & 0 & 39 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 39 & 0 & 0 & 9 \end{pmatrix}.$$

Direct calculation gives

$$\mathbf{a} = \frac{88}{97}\mathbf{e}_z, \qquad \mathbf{b} = \frac{72}{97}\mathbf{e}_z, \qquad T = \operatorname{diag}\left(\frac{39}{97}, -\frac{39}{97}, \frac{81}{97}\right), \qquad F_3(\rho)^2 = \frac{99}{97} > 1.$$

The actual output centre is still 16/25. This value attains the global centre-constrained envelope and the probability-constrained envelope at $p = 97/169$; the filter saturates its largest allowed singular value. At this same input centre, strict activation with success floor p_0 is possible exactly when $p_0 < 97/144$. Thus the optimum-correlation protocol and the largest permissible success floor answer distinct quantitative questions. The exact rational script `calculations/exact_physical_example.py` verifies the input and output principal minors, both marginal vectors, the centre identity, the success probability, and the two frontier values.

7 Sharp volume bounds at fixed centre, radial width, and correlation

The determinant of the canonical correlation matrix controls the steering ellipsoid volume: $\mathrm{Vol}(\mathcal{E}) = (4\pi/3)|\det S|$. The preceding centre/radial correlation theorem admits a distinct sharp volume refinement. In particular, the state maximizing correlation at a fixed radial width generally does not maximize volume at that width.

Theorem 11 (Exact signed-determinant interval)**.** *For canonical data $(c\mathbf{n}, 0, S)$ with $\|\mathbf{n}\| = 1$, $0 \le c \le 1$ and $t = \left\|S^{\mathsf{T}}\mathbf{n}\right\| \in [0, 1-c]$, the exact signed-determinant range is*

$$-\frac{t((1+t)^2 - c^2)}{4} \;\le\; \det S \;\le\; \frac{t((1-t)^2 - c^2)}{4}. \tag{36}$$

Every value in this interval is attained. Consequently the sharp absolute volume bound is

$$\boxed{|\det S| \le \frac{t((1+t)^2 - c^2)}{4}.} \tag{37}$$

For $t > 0$ the absolute maximum has a unique local-unitary orbit. In radial axes it is (24) with the 00–11 coherence $\sqrt{ad}$ and the 01–10 coherence zero. Its rank is three if $t < 1 - c$, two if $0 < c < 1$ and $t = 1 - c$, and one at $(c, t) = (0, 1)$. At $t = 0$, every feasible state has determinant zero.

Proof. Rotate into the radial axes used in Theorem 7. The state then has the form

$$\sigma = \begin{pmatrix} a & 0 & z & w \\ 0 & b & v & -z \\ \overline{z} & \overline{v} & e & 0 \\ \overline{w} & -\overline{z} & 0 & d \end{pmatrix}, \tag{38}$$

where a, b, e, d have the definitions in that theorem. Its actual correlation matrix is

$$S = \begin{pmatrix} 2\Re(w+v) & 2\Im(v-w) & 4\Re z \\ -2\Im(w+v) & 2\Re(v-w) & -4\Im z \\ 0 & 0 & t \end{pmatrix}.$$

Thus, putting $W = |w|^2$, $J = |v|^2$, $Z = |z|^2$,

$$\det S = 4t(J - W), \qquad \tau = t^2 + 8(W + J + 2Z). \tag{39}$$

The two principal minors give $W \le ad$ and $J \le be$. Since $ad - be = t/4 \ge 0$, these give precisely (36) and (37). Take $z = 0$ and independently choose $0 \le W \le ad$, $0 \le J \le be$ with real nonnegative coherences. The two 2×2 blocks are positive, so every signed determinant in the claimed interval is attained.

If $t > 0$, then $ad > be$. Absolute equality forces $W = ad$ and $J = 0$. The saturated $00, 11$ principal minor and positivity force its two rows to be proportional, which forces $z = 0$. A phase rotation

removes the phase of w. The state is consequently the indicated unique orbit. Its block ranks give the stated ranks. For $t = 0$, the last row of S vanishes and the determinant is identically zero, so no uniqueness is asserted. □

At the extreme radial width $t = 1-c$, (37) reduces to $|\det S| \le (1-c)^2$, the established fixed-centre maximum of Milne et al. [5]. The refinement here fixes the radial width as well, and then also fixes the correlation strength. At smaller radial widths, the volume maximizer usually has rank three, whereas the correlation maximizer has rank two. There is a sharp quantitative tradeoff between these two objectives.

Theorem 12 (Sharp correlation–volume tradeoff)**.** *Fix any feasible c, t and any $\tau \in [t^2, 1 - c^2 + 2t^2]$. Among canonical states with these three prescribed parameters, the exact greatest absolute determinant is*

$$\boxed{V_{\max}(c,t,\tau) = \frac{t}{2}\min\{\tau - t^2,\ 1 + 2t + 2t^2 - c^2 - \tau\}.} \tag{40}$$

In particular the largest volume occurs at $\tau = t^2 + [(1+t)^2 - c^2]/2$, while at the largest correlation $\tau = 1 - c^2 + 2t^2$ the determinant magnitude is at most t^2. Every value on the upper tradeoff curve is attained by an explicit positive X state, including all rank-deficient and zero-width cases.

Proof. Use (38) and set

$$U = ad = \frac{(1+t)^2 - c^2}{16}, \quad V = be = \frac{(1-t)^2 - c^2}{16}, \quad s = \frac{\tau - t^2}{8} = W + J + 2Z.$$

The allowed correlation interval gives $0 \le s \le U + V$. The block matrices $A = \mathrm{diag}(a, b)$ and $D = \mathrm{diag}(e, d)$ satisfy $A \preceq a\mathbb{I}$ and $D \preceq d\mathbb{I}$. Block positivity gives, for arbitrary $\xi, \eta \in \mathbb{C}^2$,

$$|\xi^\dagger C \eta|^2 \le (\xi^\dagger A \xi)(\eta^\dagger D \eta) \le ad\,\|\xi\|^2\,\|\eta\|^2, \qquad C = \begin{pmatrix} z & w \\ v & -z \end{pmatrix}.$$

Here the scalar Cauchy–Schwarz inequality follows by restricting the positive block matrix to the two vectors $(\xi, 0)$ and $(0, \eta)$; it holds when either restricted diagonal entry is zero. Consequently $\|C\|_{\mathrm{op}}^2 \le U$, and in particular

$$W + Z \le U, \qquad J + Z \le U.$$

No inverse, strict positivity assumption, or limiting argument is needed. The elementary inequalities

$$|W - J| \le W + J \le s, \qquad |W - J| = 2\max\{W, J\} + 2Z - s \le 2U - s$$

therefore give $|\det S| \le 4t \min\{s, 2U - s\}$. Substituting the definitions yields (40).

To attain the bound, set $z = 0$ and take real nonnegative w, v. For $0 \le s \le U$, choose $W = s$, $J = 0$. For $U \le s \le U + V$, choose $W = U$, $J = s - U$. In each case $W \le U$ and $J \le V$, so both independent 2×2 blocks of the state are positive. The correlation is exactly $t^2 + 8s = \tau$. In the first branch the determinant magnitude is $4ts$; in the second it is $4t(2U - s)$, where $2U - s \ge U - V = t/4 \ge 0$. This proves sharpness for every feasible triple. □

Complete equality classification for the tradeoff. For $t > 0$, keep $U = ad$, $V = be$, $s = (\tau - t^2)/8$, so that $U > V$. If $0 \le s \le U$, every maximizer, after removing coherence phases by local z rotations, is one of

$$(z, w, v) = (0, \sqrt{s}, 0), \qquad (z, w, v) = (0, 0, \sqrt{s}) \quad \text{when } s \le V.$$

The two representatives coincide at $s = 0$. For $s > 0$ they have opposite signed determinant and hence belong to distinct local-unitary orbits. If $U < s \le U + V$, there is exactly one maximizing orbit, represented by

$$(z, w, v) = (0, \sqrt{U}, \sqrt{s - U}).$$

To verify necessity in the first range, equality in $|W - J| \le W + J + 2Z$ forces $Z = 0$ and $\min\{W, J\} = 0$. In the second range, equality forces $\max\{W, J\} + Z = U$. The second row actually satisfies $J + Z \le bd < ad = U$, so the maximum must be W. Applying positivity to the explicit vector $(d, 0, -\overline{z}, -\overline{w})$ gives

$$0 \le d\,[ad - W - Z] - \frac{t}{2}Z.$$

Since $t > 0$, equality $W + Z = U$ forces $Z = 0$ and then $W = U$. The displayed phase removal uses two independent local phase rotations, so it preserves all physical parameters and the signed determinant. At $t = 0$ every feasible state has zero volume and is a maximizer.

When $t < 1 - c$ and $t > 0$, the first, negative-chirality branch has rank four for $s < U$, rank three at $s = U$, rank three for $U < s < U + V$, and rank two at $s = U + V$. The secondary positive-chirality branch has rank four for $s < V$ and rank three at $s = V$. Zero diagonal entries at $t = 1 - c$ reduce these ranks by the same block-counting rule used above.

The alignment of the two qubits used in these written proofs is realized by local qubit unitaries. Signed-determinant invariance can be verified directly, without assuming a correspondence with every real orthogonal matrix. Using $\sigma_i\sigma_j = \delta_{ij}\mathbb{I} + i\epsilon_{ijk}\sigma_k$, expansion of a canonical density matrix gives

$$16\,\mathrm{Tr}(\sigma^3) = 1 + 3c^2 + 3\tau - 6\det S.$$

Here $c^2 = 2\,\mathrm{Tr}(\sigma_A^2) - 1$ and $\tau = 4\,\mathrm{Tr}(\sigma^2) - 1 - c^2$. A local unitary preserves each of these three traces and keeps Bob maximally mixed. It therefore preserves $\det S$ itself. This proves the asserted distinction between the two signed orbits, and explains why the formulas describe chirality as well as volume.

Formalization boundary. The finite-matrix determinant identity, both signed determinant bounds, the rank-safe row quadratic-form tests, and the three-parameter upper bound are checked in `MilneVolumeGeometry.lean`; the principal statement is `alignedState_triple_volume_bound`. The two independent positive coherence families and exact attained maximum are checked in `MilneVolumeSharpness.lean`, principally by `alignedTripleVolumeSpectrum_isGreatest` and `volumeWitness_attains`.

The final results are stated for arbitrary density matrices in `MilneVolumePhysical.lean`, principally by `physicalTripleVolumeSpectrum_isGreatest` and `physicalRadialVolumeSpectrum_isGreatest`. Their domain imposes the actual Pauli conditions $\mathbf{a} = c\mathbf{e}_z$, $\mathbf{b} = 0$, and $\left\|S^{\mathsf{T}}\mathbf{e}_z\right\|^2 = t^2$, and imposes no form on the other matrix entries. The axis $\mathbf{e}_z$ is a fixed marked axis, retained also when $c = 0$. The checked Bob-only normalization in `MilneBobRadialNormalForm.lean` puts every such density matrix into radial coordinates while preserving that marked Alice axis. `MilneVolumeUnitary.lean` verifies the cubic-trace identity and the signed-determinant invariance under actual local unitaries. Thus no positivity surrogate, assumed row bound, assumed unitary invariance, or rank assumption is supplied to the final physical theorems. The additional orbit equality classification, complete signed-interval attainability, and detailed rank counts in this section are written proofs; they are not claimed as separately formalized Lean theorems.

8 Complete equality classification

Appendix A gives the constructive unitary and matrix proofs behind the formal classification, including repeated eigenvalues, vanishing coherences, and the pure-product endpoint.

Theorem 13 (Extremizing states)**.** *For $0 < c < 2/3$, every state attaining $G(c)$ is locally unitarily equivalent to (17) with $r = r_*(c)$. At $c = 0$, the maximizers are exactly the maximally entangled*

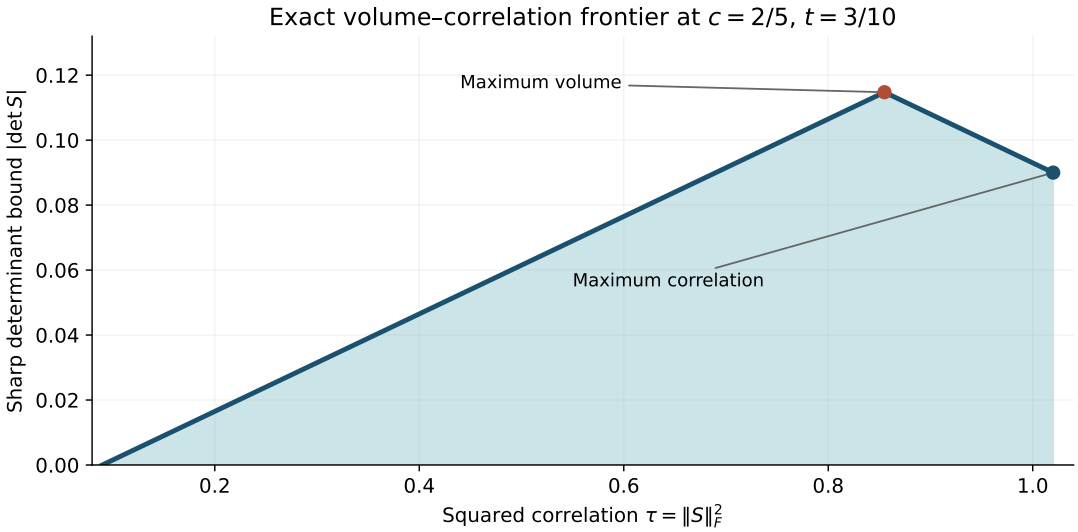


Figure 2: The exact determinant–correlation frontier at $c = 2/5$, $t = 3/10$. The feasible correlation interval is $[0.09, 1.02]$. The maximum volume occurs at $\tau = 0.855$, with $|\det S| = 0.11475$; at maximum correlation, $|\det S| = 0.09$. Each boundary value has an explicit positive density-matrix witness, including all singular endpoints.

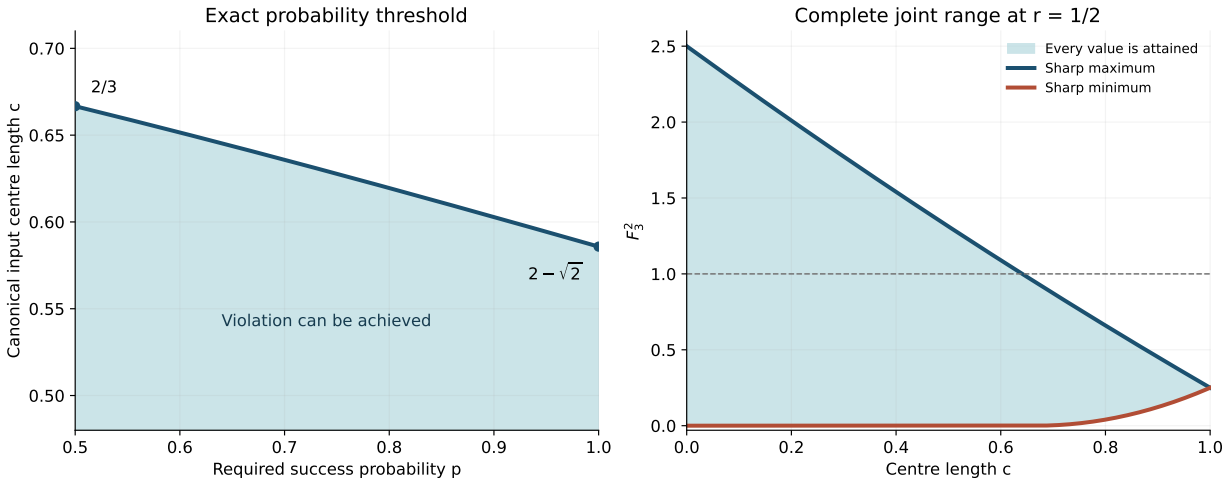


Figure 3: Two exact refinements. Left: the strict activation region lies below $C_{\mathrm{crit}}(p)$, optimizing over canonical inputs at each centre. Right: every point between the sharp lower and upper curves is attained at Bob Bloch length $r = 1/2$. The dashed line is the $F_3^2 = 1$ threshold.

pure states. There are no maximizers for $2/3 \le c < 1$. *At* $c = 1$, *the maximizers are exactly pure product states.*

Proof. First classify canonical equality in $\operatorname{tr} Q \le (1-c)(3-c)$. Rotate Alice so $\mathbf{c} = c\mathbf{e}_z$, and Bob so $S^{\mathsf{T}}\mathbf{e}_z = t\mathbf{e}_z$, $t \ge 0$. At $c = 0$, choose any Alice direction before this rotation. The general canonical matrix now has form

$$\sigma = \begin{pmatrix} a_+ & 0 & z & w \\ 0 & a_- & v & -z \\ \overline{z} & \overline{v} & d_- & 0 \\ \overline{w} & -\overline{z} & 0 & d_+ \end{pmatrix}. \tag{41}$$

The diagonal Bob-valued blocks sum to $\mathbb{I}/2$, commute, and were simultaneously diagonalized by the Bob rotation. The zero Alice transverse Bloch components give the $z, -z$ constraint. Direct computation yields

$$\operatorname{tr} Q = t^2 + 16|z|^2 + 8|w|^2 + 8|v|^2. \tag{42}$$

Canonical equality and (8) force $t = 1 - c$. Then $d_- = 0$. Positivity of 2×2 principal minors forces the entire third row and column to vanish, so $z = v = 0$. Equation (42) forces $|w| = \sqrt{1-c}/2$. A local Z rotation makes w nonnegative, yielding exactly $\sigma_{c,1-c}$. This argument includes all zero diagonal entries. At $c = 1$, the canonical state is pure on Alice and maximally mixed on Bob.

For $0 < c < 2/3$, (15) and the nonnegative losses in Section 9 force $r = r_*$, canonical equality, and equality in (11). Here $0 < r_* < 1$, so equality in the triangle inequality requires the conditional Bloch vector $\mathbf{y} = \mathbf{c} + S\widehat{\mathbf{b}}$ to equal $\mathbf{e}_z$. In the aligned obese form $S = \operatorname{diag}(\sqrt{1-c}, -\sqrt{1-c}, 1-c)$ this forces $\widehat{\mathbf{b}} = \mathbf{e}_z$. The state is therefore (17) with its unique filter length. For $c = 0$, the square term forces $r = 0$ and the canonical equality class is the Bell orbit. The flat and endpoint cases follow from Theorem 4 and the product calculation at $r = 1$. □

At fixed (c, r) with $r < 1$, the same reasoning classifies equality in (13). For $0 < c < 1$, the canonical state must be maximally obese; when $r > 0$ the filter points toward its outward contact with the Bloch sphere. At $r = 0$ there is no filter direction. At $c = 0$ equality consists of pure states with the specified marginal length r; at $c = 1$ it consists of a pure Alice state times a Bob state of length r. At $r = 1$ every state is a product and $F_3^2 = c^2$.

9 Exact losses and quantitative rigidity

The following is stronger than the original centre-only estimate because it records separately where each loss occurs.

Proposition 14 (Exact deficit identity)**.** *For* $0 \le c < 2/3$ *and* $r < 1$, *put* $\delta = 3 - 4c + c^2 - \|S\|_F^2$. *Then*

$$G(c) - F_3^2 = D(c)(r - r_*)^2 + (1 - r^2)\delta + [c + (1-c)r]^2 - \|\mathbf{a}\|^2. \tag{43}$$

All three terms are nonnegative. Thus equality is equivalent to the simultaneous vanishing of these three explicitly defined matrix losses. If $c > 0$, *this implies the sharper geometric inequality*

$$G(c) - F_3^2 \ge D(c)(r - r_*)^2 + 2(1 - r^2)((1-c)^2 - q). \tag{44}$$

Proof. Subtract (10) from (15) together with the definition of $H_c(r)$. Nonnegativity follows from the canonical and marginal bounds. Then apply (8). □

There is a more detailed identity. For $0 < c < 2/3$ and $0 < r < 1$, set $\mathbf{n} = \mathbf{c}/c$, $\mathbf{y} = \mathbf{c} + S\widehat{\mathbf{b}}$, and $\tau = 4\,\mathrm{Tr}(\sigma\sigma_{\mathbf{n}}) = 1 - c^2 + 2q - \mathrm{tr}\,Q$. Then

$$\begin{aligned} G(c) - F_3^2 = D(c)(r - r_*)^2 + (1 - r^2)\{2[(1-c)^2 - q] + \tau\} \\ + r[r + c(1-r)](1 - \|\mathbf{y}\|^2) + rc(1-r)\,\|\mathbf{y} - \mathbf{n}\|^2 . \end{aligned} \tag{45}$$

Each displayed contribution is nonnegative. The last line follows by expanding $\mathbf{a} = (1-r)c\mathbf{n} + r\mathbf{y}$ and using $2(1 - \mathbf{n}\cdot\mathbf{y}) = 1 - \|\mathbf{y}\|^2 + \|\mathbf{y} - \mathbf{n}\|^2$. At $r = 0$ the endpoint terms vanish; at $c = 0$ use (43) without dividing by c. The five losses correspond to filter length, radial support width, spin-flip overlap, conditional-state impurity, and endpoint direction.

Theorem 15 (Explicit stability of the canonical optimizer)**.** *For a canonical state with $0 \le c < 1$, put $d = 1 - c$, let t be the radial support half-width, and put $\delta = 3 - 4c + c^2 - \mathrm{tr}\,Q$. There exist local unitaries U_A, U_B such that, with $U = U_A \otimes U_B$,*

$$\begin{aligned} \left\| U\sigma U^\dagger - \sigma_{c,d} \right\|_1^2 &\le \min\left\{4, \left[\frac{1}{2} + \frac{2+c}{d+t}\right]\delta\right\} \\ &\le \min\left\{4, \frac{5+c}{2(1-c)}\delta\right\}. \end{aligned} \tag{46}$$

The norm is the trace norm. For the convention that trace distance is one half of that norm, divide the square-root bounds by two. The constants are proved upper bounds and are not claimed optimal.

Proof. Use (41); make $w \ge 0$ by a local phase rotation. Let $y = d - t \in [0, d]$ and $w_0 = \sqrt{d}/2$. Principal-minor positivity gives

$$w^2 \le a_+ d_+ \le w_0^2, \qquad |z|^2 \le a_+ y/4, \qquad |v|^2 \le a_- y/4.$$

In particular,

$$16(2|z|^2 + |v|^2) \le y(3 + 3c + t) = y(4 + 2c - y). \tag{47}$$

Equation (42) implies exactly

$$\delta = (d^2 - t^2) + 8(w_0^2 - w^2) - 16|z|^2 - 8|v|^2.$$

Since $0 \le w \le w_0$, $(w_0 - w)^2 \le w_0^2 - w^2$. The four diagonal differences from $\sigma_{c,d}$ are $\pm y/4$, so

$$\|\sigma - \sigma_{c,d}\|_F^2 = y^2/4 + 4|z|^2 + 2|v|^2 + 2(w_0 - w)^2.$$

For 4×4 matrices, Cauchy–Schwarz on singular values gives $\|X\|_1^2 \le 4\,\|X\|_F^2$. Retaining the radial deficit term yields

$$\begin{aligned} \|\sigma - \sigma_{c,d}\|_1^2 &\le \delta + y^2 - (d^2 - t^2) + 16(2|z|^2 + |v|^2) \\ &\le \delta + y(3 + 3c - t). \end{aligned}$$

By (8), $\delta \ge 2y(d + t)$. Since $d + t > 0$ and $3 + 3c - t \ge 0$, we obtain

$$\|\sigma - \sigma_{c,d}\|_1^2 \le \delta\left[1 + \frac{3 + 3c - t}{2(d+t)}\right] = \delta\left[\frac{1}{2} + \frac{2+c}{d+t}\right].$$

Use $d + t \ge d$ for the uniform version and $\|\sigma - \sigma_{c,d}\|_1 \le 2$ for the other branch. None of the arguments divides by a diagonal entry or assumes a full-rank state. □

In particular, if $F_3(\rho)^2 \geq G(c) - \varepsilon$, $c < 2/3$, and $r < 1$, then

$$|r - r_*| \leq \sqrt{\varepsilon / D(c)}, \qquad \delta \leq \frac{\varepsilon}{1 - r^2}.$$

The whitened state is therefore, up to local unitaries, within trace norm

$$\min\left\{2, \sqrt{\frac{(5+c)\varepsilon}{2(1-c)(1-r^2)}}\right\}$$

of the canonical obese optimizer. This conclusion concerns the whitened state; it is not a claimed trace-distance bound to the globally optimal *unfiltered* state.

9.1 Extremizing sequences and a sharp change in stability exponent

The unattained branch has a different stability phenomenon from the curved branch. Its limiting pure products have centre length one, even when every state in the sequence has the same smaller centre length.

Theorem 16 (Sharp rates on the unattained branch)**.** *Fix $2/3 \leq c < 1$. If a sequence of states with centre length c has $F_3(\rho_j)^2 \to 1$, then eventually its Bob marginals are full rank, $r_j \to 1$, and its distance to the set $\mathcal{P}$ of pure product states tends to zero. Put*

$$\varepsilon = 1 - F_3(\rho)^2, \quad h = 1 - r, \quad A = 4c - 2 - c^2, \quad B = 6c - 4, \quad D = 2 - 2c - c^2.$$

For $r < 1$,

$$\varepsilon \geq Ah^2 + B(1-h)h = Bh + Dh^2, \qquad \inf_{\pi \in \mathcal{P}} \|\rho - \pi\|_1^2 \leq 2\varepsilon + 14h. \tag{48}$$

Consequently, at fixed c, as $\varepsilon \downarrow 0$,

$$\inf_{\pi \in \mathcal{P}} \|\rho - \pi\|_1 = \begin{cases} O(\varepsilon^{1/4}), & c = 2/3, \\ O(\varepsilon^{1/2}), & 2/3 < c < 1. \end{cases} \tag{49}$$

Both exponents are sharp: neither can be increased in a bound valid for all states at the specified fixed centre length.

Proof. A state with $r = 1$ has $F_3^2 = c^2 < 1$, so such states cannot occur arbitrarily late in an extremizing sequence. Equation (16) gives the first bound in (48). Since $A > 0$ and $B \geq 0$, it forces $h \to 0$. The exact filter identity gives

$$1 - \|\mathbf{a}\|^2 = \varepsilon + (1 - r^2)(\|S\|_F^2 - c^2) \leq \varepsilon + 3(1 - r^2) \leq \varepsilon + 6h.$$

Choose the top eigenvectors of both marginal states and let π be their product projector. The elementary union bound for their commuting projectors gives $\mathrm{Tr}(\rho\pi) \geq (\|\mathbf{a}\| + r)/2$. The pure-target trace-norm bound $\|\rho - \pi\|_1 \leq 2\sqrt{1 - \mathrm{Tr}(\rho\pi)}$ follows, for example, by spectral decomposition, the pure-state formula, and concavity of the square root. Since $1 - \|\mathbf{a}\| \leq 1 - \|\mathbf{a}\|^2$, this proves the second inequality in (48).

At $c = 2/3$, $B = 0$ and $D = 2/9$, so $h \leq 3\sqrt{\varepsilon}/\sqrt{2}$. At fixed $c > 2/3$, once $h \leq 1/2$, the first expression in (48) gives $h \leq 2\varepsilon/B$. Substitution proves (49).

For sharpness, take $\rho_{c,1-h}$ from (17). It has exactly $\varepsilon = Bh + Dh^2$. Let $W = (|\psi^-\rangle\langle\psi^-|)^{\Gamma_B}$, with $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$. This operator has operator norm $1/2$ and nonnegative expectation in every separable state, because its expectation in a product is a squared overlap with a partially conjugated product vector. Direct calculation gives

$$\mathrm{Tr}(W\rho_{c,1-h}) = \frac{ch}{4} - \frac{1}{2}\sqrt{(1-c)(2h - h^2)}.$$

Trace-norm duality therefore implies, even for the distance to the larger set of separable states,

$$\inf_{\pi\in\mathcal{P}} \|\rho_{c,1-h} - \pi\|_1 \geq \sqrt{(1-c)(2h-h^2)} - \frac{ch}{2}. \tag{50}$$

Its leading term is $\sqrt{2(1-c)h}$. At $c = 2/3$ the leading lower bound is $2^{1/4}\varepsilon^{1/4}$; at fixed $2/3 < c < 1$ it is $\sqrt{2(1-c)/(6c-4)}\sqrt{\varepsilon}$. This proves sharpness of both exponents. The constants are not asserted uniform in c, and the case $c = 1$ is the separately attained endpoint. □

For a *fixed* canonical state with $2/3 \leq c < 1$, the supremum over invertible Bob filters equals one exactly when its ellipsoid has a pure contact: $\|\mathbf{c} + S\mathbf{u}\| = 1$ for some unit $\mathbf{u}$. Indeed, the continuous objective on the closed filter ball has no interior point of value one, and on the sphere it is exactly $\|\mathbf{c} + S\mathbf{u}\|^2$. Compactness proves necessity; approaching a pure contact radially proves sufficiency. For any sequence in this fixed orbit with $F_3^2 \to 1$, every cluster point of its filter directions lies in this contact set. Thus an individual ellipsoid can have an orbit supremum strictly below the global value one.

10 An exact test for one-sided hidden F_3 violation

The centre bound is a universal obstruction. Using the entire ellipsoid gives a necessary and sufficient test for whether any Bob-only filter can reveal $F_3 > 1$. This directly addresses the asymmetric filtering problem raised in Eqs. (9)–(10) and the conclusion of the 2024 paper [1].

Fix canonical Bloch data $(\mathbf{c}, 0, S)$ and put

$$\tau = \|S\|_F^2, \qquad \mathbf{k} = S^{\mathsf{T}}\mathbf{c}, \qquad K = (\tau - c^2)\mathbb{I} - S^{\mathsf{T}}S. \tag{51}$$

Every positive invertible Bob filter, after normalization, is represented by $F = \sqrt{\mathbb{I} + \mathbf{b}\cdot\boldsymbol{\sigma}}$, $\|\mathbf{b}\| < 1$. Polar decomposition shows that post-filter unitaries do not enlarge the achievable F_3 values. From (10) and $\mathbf{a} = \mathbf{c} + S\mathbf{b}$, the exact squared value is

$$f(\mathbf{b}) = \tau + 2\mathbf{k}\cdot\mathbf{b} - \mathbf{b}^{\mathsf{T}}K\mathbf{b}. \tag{52}$$

On the unit sphere its continuous boundary value is

$$f(\mathbf{b}) = \|\mathbf{c} + S\mathbf{b}\|^2 \leq 1. \tag{53}$$

Boundary parameters represent singular filters producing product states.

Theorem 17 (One-sided violation criterion and exact optimum)**.** *Some invertible Bob filter produces $F_3 > 1$ if and only if*

$$K \succ 0, \qquad \left\|K^{-1}\mathbf{k}\right\|^2 < 1, \qquad V := \tau + \mathbf{k}^{\mathsf{T}}K^{-1}\mathbf{k} > 1. \tag{54}$$

When these conditions hold, the maximum squared value is V, attained at the unique positive-filter parameter $\mathbf{b}_* = K^{-1}\mathbf{k}$. Singular or indefinite K cannot permit an F_3 violation.

Proof. A value greater than one in the open ball gives, by continuity, a maximum greater than one on the compact closed ball. Equation (53) forces it to be interior. First and second derivative conditions give $K\mathbf{b}_* = \mathbf{k}$ and $K \succeq 0$. If K had a nonzero null vector $\mathbf{v}$, then $\mathbf{k}\cdot\mathbf{v} = \mathbf{b}_*^{\mathsf{T}}K\mathbf{v} = 0$. The value of f would be constant on the line $\mathbf{b}_* + s\mathbf{v}$, which meets the unit sphere, contradicting (53). Thus $K \succ 0$. Completing the square gives

$$f(\mathbf{b}) = V - (\mathbf{b} - \mathbf{b}_*)^{\mathsf{T}}K(\mathbf{b} - \mathbf{b}_*).$$

This proves the necessary conditions, value, and uniqueness. Conversely, (54) places this stationary point in the physical open ball with value greater than one; the same identity proves optimality. □

For an original state with full-rank Bob marginal, whitening followed by this filter is one Bob filter, scaled to a valid Kraus contraction. Conversely every Bob filter is represented in this canonical orbit up to a Bob unitary. A pure Bob marginal is a product case and never violates. Thus both possible marginal ranks and every global rank are covered.

10.1 Ellipsoid-only and exact polynomial forms

Put $L = (\tau - c^2)\mathbb{I} - Q$, where $Q = SS^{\mathsf{T}}$. The matrices K, L have the same eigenvalues, including when S is singular. Intertwining yields

$$L \succ 0, \qquad \mathbf{c}^{\mathsf{T}} Q L^{-2} \mathbf{c} < 1, \qquad \tau + \mathbf{c}^{\mathsf{T}} Q L^{-1} \mathbf{c} > 1. \tag{55}$$

This equivalent test depends only on the centre and ellipsoid matrix. For rational data, let $\ell = \det L$ and $R = \operatorname{adj} L$. After certifying $L \succ 0$ by principal minors, so $\ell > 0$, the other tests are

$$\mathbf{c}^{\mathsf{T}} Q R^2 \mathbf{c} < \ell^2, \qquad (\tau - 1)\ell + \mathbf{c}^{\mathsf{T}} Q R \mathbf{c} > 0.$$

These are finite polynomial inequalities and require no numerical inverse.

The unit-ball condition cannot be dropped. The positive definite canonical state

$$\sigma = \frac{1}{40} \begin{pmatrix} 19 & 0 & 0 & 5 \\ 0 & 7 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 5 & 0 & 0 & 13 \end{pmatrix}$$

has two positive 2×2 blocks with determinants $222/1600$ and $6/1600$. Its data are $\mathbf{c} = (0, 0, 3/10)$, $S = \operatorname{diag}(3/10, -1/5, 3/5)$, $\tau = 49/100$, $K = \operatorname{diag}(31, 36, 4)/100 \succ 0$. The unconstrained vertex is $(0, 0, 9/2)$ with value $13/10 > 1$, outside the physical ball. The actual supremum is $81/100 < 1$, from

$$f(x, y, z) = \frac{49}{100} + \frac{9}{25} z - \frac{31}{100} x^2 - \frac{9}{25} y^2 - \frac{1}{25} z^2.$$

It increases with z on $[-1, 1]$, and its transverse terms are nonpositive.

The criterion gives another proof of the threshold. For $c > 0$ and $q = \mathbf{n}^{\mathsf{T}} Q \mathbf{n}$, $\mathbf{n} = \mathbf{c}/c$, some direction perpendicular to $\mathbf{n}$ has Rayleigh quotient at least $(\tau - q)/2$. Hence

$$\lambda_{\min}(L) \leq \frac{\tau + q - 2c^2}{2} \leq \frac{1 - 3c^2 + 3q}{2} \leq 2 - 3c.$$

For $c \geq 2/3$, the necessary positive-definiteness condition fails.

10.2 The entire orbit supremum, including singular quadratic cases

For completeness, the exact scalar trust-region characterization is

$$\sup_{\|\mathbf{b}\| < 1} f(\mathbf{b}) = \tau + \min_{\substack{\lambda \geq 0,\ K + \lambda \mathbb{I} \succeq 0 \\ \mathbf{k} \in \operatorname{range}(K + \lambda \mathbb{I})}} \left[\lambda + \mathbf{k}^{\mathsf{T}} (K + \lambda \mathbb{I})^{\dagger} \mathbf{k} \right]. \tag{56}$$

Here $\dagger$ is the Moore–Penrose inverse. For the upper bound, put $B = K + \lambda \mathbb{I}$ and complete the square:

$$f(\mathbf{b}) = \tau + \lambda \|\mathbf{b}\|^2 + \mathbf{k}^{\mathsf{T}} B^{\dagger} \mathbf{k} - (\mathbf{b} - B^{\dagger} \mathbf{k})^{\mathsf{T}} B (\mathbf{b} - B^{\dagger} \mathbf{k}).$$

For equality, use $\lambda = 0$ if $K \succeq 0$, $\mathbf{k} \in \operatorname{range} K$, and $\left\| K^{\dagger} \mathbf{k} \right\| \leq 1$. Otherwise let $\lambda_0 = \max(0, -\lambda_{\min}(K))$. For $\lambda > \lambda_0$, $\left\| (K + \lambda \mathbb{I})^{-1} \mathbf{k} \right\|$ is continuous, nonincreasing, and tends to zero. If its limit at λ_0 exceeds one, choose a root with norm one. If the limit is at most one, the

excluded $\lambda = 0$ case implies $\lambda_0 > 0$; the limiting matrix is singular and $\mathbf{k}$ is orthogonal to its kernel. Add a kernel vector to its minimum-norm solution to reach the unit sphere. The square term vanishes in both cases, giving equality. The open ball has the same supremum as its closure by continuity, but need not attain a boundary optimum.

In a K eigenbasis, the scalar objective is $\lambda + \sum_i k_i^2/(\kappa_i + \lambda)$; a zero denominator is admissible only when the corresponding numerator vanishes. Unlike the formula in the violating regime, this exact characterization retains a scalar minimization.

11 Complete one-sided orbit optimization, including all hard cases

The generic optimization method in this section is classical trust-region theory; see Moré–Sorensen (1983) and Fortin–Wolkowicz (2004). The extension here is its exact application to the complete physical one-sided F_3 orbit, with singular filters retained, together with a constructive formal proof. In particular, a scalar minimization is not left as the answer: an explicitly tested spectral endpoint or one uniquely admissible root of a degree-six polynomial determines the result.

11.1 Spectral algorithm for an arbitrary real quadratic

Let $K = K^T \in \mathbb{R}^{3\times 3}$, $k \in \mathbb{R}^3$, and $\tau \in \mathbb{R}$. We seek the exact maximum of

$$f(x) = \tau + 2k^T x - x^T K x, \qquad \|x\| \le 1. \tag{57}$$

Choose an orthogonal diagonalization

$$K = U\,\mathrm{diag}(d_1, d_2, d_3)U^T, \qquad h = U^T k, \qquad d_{\min} = \min_i d_i, \qquad L = \max\{0, -d_{\min}\}.$$

For $\ell \ge L$, define the partial reciprocal vector and its squared norm by

$$q_i(\ell) = \begin{cases} h_i/(d_i + \ell), & d_i + \ell \ne 0, \\ 0, & d_i + \ell = 0, \end{cases} \qquad \psi(\ell) = \sum_i q_i(\ell)^2.$$

The zero-denominator convention alone does not make the endpoint admissible: the range condition below must also hold.

Exact finite algorithm. First test the endpoint condition

$$h_i = 0 \quad \text{whenever } d_i + L = 0, \qquad \psi(L) \le 1. \tag{58}$$

If it holds, set $\ell_* = L$. If it fails, let $\ell_* > L$ be the unique solution of

$$\sum_{i=1}^{3} \frac{h_i^2}{(d_i + \ell_*)^2} = 1. \tag{59}$$

This regular equation is equivalently the polynomial equation

$$\begin{aligned} P(\ell) = {} & h_1^2(d_2+\ell)^2(d_3+\ell)^2 + h_2^2(d_1+\ell)^2(d_3+\ell)^2 \\ & + h_3^2(d_1+\ell)^2(d_2+\ell)^2 - (d_1+\ell)^2(d_2+\ell)^2(d_3+\ell)^2 = 0. \end{aligned} \tag{60}$$

Its degree is exactly six, with leading coefficient -1, including when $h = 0$ or eigenvalues coincide. Consequently the endpoint and the real roots of P form a set of at most seven candidates. The

endpoint test (58) and the regular restriction $\ell > L$ select exactly one admissible candidate. Clearing denominators introduces no spurious candidate on that regular interval, because every denominator is strictly positive there. In the original coordinates the same regular polynomial equation can be written

$$\| \operatorname{adj}(K + \ell I)k\|^2 - \det(K + \ell I)^2 = 0.$$

The spectral version is particularly convenient for explicitly resolving the singular endpoint.

Exact maximum and every maximizing point. The answer is

$$M = \tau + \ell_* + \sum_{i:\, d_i + \ell_* \neq 0} \frac{h_i^2}{d_i + \ell_*}. \tag{61}$$

Put $q = q(\ell_*)$ and $E_* = \{z \in \mathbb{R}^3 : z_i = 0 \text{ whenever } d_i + \ell_* \neq 0\}$. The complete maximizing set is

$$\left\{ U(q + z) : z \in E_*, \quad \begin{cases} \|z\|^2 \leq 1 - \psi(\ell_*), & \ell_* = 0, \\ \|z\|^2 = 1 - \psi(\ell_*), & \ell_* > 0. \end{cases} \right\}. \tag{62}$$

The endpoint range condition guarantees $h_i = 0$ on E_*, and q is orthogonal to E_*. Thus this description includes every degenerate case:

- If $\ell_* > L$, all shifted eigenvalues are positive, and the unique maximizer is Uq, on the unit sphere.
- If $\ell_* = 0$, the original K is positive semidefinite. The maximizers form the explicitly displayed affine ball in $\ker K$. For $K \succ 0$ this is the single unconstrained vertex $K^{-1}k$.
- If $\ell_* = L > 0$, then $d_{\min} < 0$ and h is orthogonal to the lowest-eigenvalue eigenspace. This is the hard case. The missing radius is supplied by a vector in that eigenspace, giving precisely the sphere in (62). When its radius is zero the maximizer is nevertheless unique.
- If $k = 0$, the formula reduces to $M = \tau + \max\{0, -d_{\min}\}$. For $d_{\min} > 0$ the maximizer is zero; for $d_{\min} = 0$ it is the unit ball in the kernel; for $d_{\min} < 0$ it is the unit sphere in the lowest-eigenvalue eigenspace. This includes $K = 0$.

11.2 Constructive proof, without assumed optimality conditions

In the eigenbasis, write $y = U^T x$ and $f_D(y) = \tau + \sum_i (2h_i y_i - d_i y_i^2)$. We construct p and ℓ satisfying

$$\ell \geq 0, \quad d_i + \ell \geq 0, \quad (d_i + \ell)p_i = h_i, \quad \|p\| \leq 1, \quad \ell(1 - \|p\|^2) = 0. \tag{63}$$

These conditions are conclusions of the construction, not hypotheses substituted for the optimization problem.

Suppose first that (58) holds. At $\ell = L = 0$, take $p = q(L)$. At $\ell = L > 0$, choose an index j for which $d_j + L = 0$, and set

$$p = q(L) + \sqrt{1 - \psi(L)}\, e_j.$$

The range condition gives $q_j(L) = h_j = 0$, so $\|p\| = 1$ and every equation in (63) holds. This directly treats the hard case, including repeated lowest eigenvalues and a zero linear term.

If the endpoint is inadmissible, a regular root exists by a finite bracket. When the range condition holds but $\psi(L) > 1$, choose $a = L$. If instead $d_j + L = 0$ and $h_j \neq 0$ for some j, choose $a = L + |h_j|/2$. The jth summand of $\psi(a)$ is then exactly 4, so $\psi(a) > 1$. In either case all

denominators at a are nonnegative, and any zero denominator has zero numerator. The function ψ is therefore continuous on $[a, \infty)$. For the explicit endpoint

$$b = a + 1 + \|h\|$$

we have

$$\psi(b) \le \frac{\|h\|^2}{(1 + \|h\|)^2} < 1.$$

The intermediate value theorem gives a root $\ell \in (a, b)$, hence $\ell > L$. Now take $p = q(\ell)$; this proves (63) in the remaining case. No compactness or differentiability assumption about an unknown optimizer was needed.

For every such constructed certificate and every $y \in \mathbb{R}^3$, direct expansion yields the exact gap identity

$$f_D(p) - f_D(y) = \sum_i (d_i + \ell)(y_i - p_i)^2 + \ell(1 - \|y\|^2). \tag{64}$$

Both terms are nonnegative on the ball, proving global optimality. Equality holds exactly when $y_i = p_i$ for every strictly positive $d_i + \ell$, together with $\ell(1 - \|y\|^2) = 0$. This proves (62). Multiplying each stationary equation by p_i and using complementarity gives (61).

For completeness, the admissible shift is also unique in the hard and flat cases. Let (p, ℓ) and (q, m) be two certificates. Each point is globally maximizing, so equality in (64) gives $(d_i + \ell)q_i = h_i$ and $\ell(1 - \|q\|^2) = 0$. Subtracting $(d_i + m)q_i = h_i$ yields $(m - \ell)q_i = 0$ for every i. If $m \neq \ell$, then $q = 0$, and the two complementarity equations force $m = \ell = 0$, a contradiction. Thus the algorithm selects one shift even when it selects a continuum of maximizing vectors.

Equivalently, in the original coordinates, a point x is globally maximizing if and only if it belongs to the unit ball and there exists $\ell \ge 0$ with

$$K + \ell I \succeq 0, \qquad (K + \ell I)x = k, \qquad \ell(1 - \|x\|^2) = 0.$$

The proof above establishes existence of this global certificate for every real symmetric K, rather than assuming this characterization.

11.3 Exact application to all canonical quantum-filter orbits

Let σ be any actual two-qubit density matrix with Bob marginal $I/2$, Alice Bloch vector $c \in \mathbb{R}^3$, and correlation matrix S. Set

$$\tau = \|S\|_F^2, \qquad k = S^T c, \qquad K = (\tau - \|c\|^2)I - S^T S. \tag{65}$$

For every complex 2×2 filter F normalized by $\mathrm{Tr}(F^\dagger F) = 2$, there is an x in the closed unit ball such that $F^\dagger F = I + x \cdot \sigma$. The actual normalized output $\rho_F = (I \otimes F)\sigma(I \otimes F^\dagger)$ satisfies the exact identity

$$F_3(\rho_F)^2 = \tau + 2k^T x - x^T K x. \tag{66}$$

Conversely every such x is realized by the positive filter $F = (I + x \cdot \sigma)^{1/2}$. These statements hold at every joint-state rank and include singular filters.

Apply the finite algorithm to (65). The exact maximum over all normalized complex one-sided filters is the number M in (61), and the exact maximum of F_3 is $\sqrt{M}$. Every maximizing effect, and no other effect, is

$$F^\dagger F = I + (U(q + z)) \cdot \sigma,$$

with z exactly as in (62). All complex filters with a given maximizing effect are included; no Hermiticity restriction is imposed on F. A positive square-root filter explicitly realizes each maximizing effect.

This completes the orbit optimization even when its maximum is at or below the steering threshold. The earlier violation criterion is recovered as the special case in which the physical sphere bound $F_3^2 \le 1$ forces an interior optimizer with $K \succ 0$.

11.4 Invertible filters: supremum versus attainment

For normalized filters,

$$\det(F^\dagger F) = 1 - \|x\|^2,$$

so invertibility is equivalent to $\|x\| < 1$. The exact supremum over invertible filters is still M: if x_* is any closed-ball maximizer, the effects tx_* with $0 \le t < 1$ are realized by invertible positive filters, and their values tend to M as $t \uparrow 1$.

An invertible filter attains that supremum if and only if

$$\ell_* = 0 \quad \text{and} \quad \psi(\ell_*) < 1. \tag{67}$$

Indeed positive shift forces every optimizer onto the unit sphere. At zero shift the squared norm of $q + z$ is $\psi(0) + \|z\|^2$, and an interior point exists exactly under (67). In particular, an endpoint maximum can be attained by singular filters while only being approached by invertible ones. This distinction is not removed by assuming that the input joint state has full rank.

11.5 State rank, quadratic rank, and filter rank are different

Two full-rank physical examples show why the singular cases cannot be removed by imposing positivity on the input state.

First take

$$\sigma = \frac{I + cZ}{2} \otimes \frac{I}{2}, \qquad 0 < c < 1.$$

Here $S = 0$, $k = 0$, and $K = -c^2 I$. The objective is $f(x) = c^2\|x\|^2$. Its maximum is c^2, every unit vector maximizes, and the spectral shift is $\ell_* = c^2$. This is a hard case with a three-dimensional nullspace of $K + \ell_* I$. Every maximizing filter is singular, and the same value is only a supremum for invertible filters, even though the input density matrix has rank four.

Next take

$$\sigma = \frac{1}{4}(I + tX \otimes X), \qquad 0 < t < 1.$$

This state also has rank four, but now $k = 0$ and $K = \mathrm{diag}(0, t^2, t^2)$. The exact maximum is t^2 and every point $x = (s, 0, 0)$ with $|s| \le 1$ is a maximizer. Thus the complete maximizing effect set contains both invertible and singular filters. These are direct specializations of the universal checked theorem. The exact rational certificate file includes both types of example.

Verification scope. The accompanying formal-coverage manifest identifies the exact Lean declarations and compiler evidence for the results above. Numerical comparisons are reproducibility checks only; no numerical result is a logical premise of a formal theorem.

12 Scope

The results concern the matrix quantity $F_3 = \|T\|_{\mathrm{F}}$ in the convention of Eq. (3) of [1]; $F_3 \le 1$ does not prove the existence of a local hidden-state model for all measurements. If Bob's measurement triad is fixed and only Alice's directions are optimized, the value can be smaller. The success probabilities refer to canonical inputs; an arbitrary original input has a separate whitening cost.

The actual centre of a singularly filtered output is handled separately from the input centre. The explicit trace-norm stability constants are upper bounds and are not claimed to be optimal.

13 Research process and AI disclosure

This is fully autonomous agentic research. The results, their proofs, the Lean development, the exact and numerical checkers and a draft report were produced by autonomous AI research agents, which took the earlier package [2] as their starting point; their records describe internal reviews carried out within the same automated workflow and do not name the system used.

Opus 5.5 by Anthropic used with Claude Code assessed the work for this release. It read the proofs of the joint interval, the canonical interval, the probability frontier and the volume bounds, compared the principal Lean statements with the mathematics, reran every checker from a copy, wrote the exact check `calculations/independent/exact_new_results.py`, added the whole-project axiom audit, ran the fresh Lean build recorded in `verification/`, and edited this manuscript from the draft report. The remaining proofs were carried over and read, but not all were rederived line by line.

Dev D. Goyal is responsible for publication. The work has not been externally peer reviewed; AI checks are not human peer review, no expert validation is claimed, and the literature comparison is bounded and does not establish priority. Appendix C states the formal scope and its limits.

A Constructive matrix proofs of rigidity and extremizer classification

The equality and stability results in this section concern actual complex density matrices and actual qubit unitaries. In particular, no unproved rotation-to-unitary lifting assertion is used. Write $\tau(\sigma) = \|T(\sigma)\|_F^2$, and use the trace norm $\|X\|_1 = \sum_j |\lambda_j(X)|$ for Hermitian X. The usual trace distance between states is $\frac{1}{2}\|\rho - \sigma\|_1$.

A.1 A complete radial normal form, including degenerate cases

For a canonical state, Bob's marginal is $I/2$. First diagonalize Alice's marginal by a qubit unitary, putting her Bloch vector at ce_z, where $c \geq 0$. In Alice blocks write $\sigma = \left(\begin{smallmatrix} A & C \\ C^\dagger & D \end{smallmatrix}\right)$. Then $A+D = I/2$. A second qubit unitary, acting only on Bob, diagonalizes the Hermitian matrix $H = A - D$, with its eigenvalues in decreasing order. If their difference is $t \geq 0$, the resulting state is exactly

$$M(c,t,z,w,v) = \begin{pmatrix} a_+ & 0 & z & w \\ 0 & a_- & v & -z \\ \overline{z} & \overline{v} & d_- & 0 \\ \overline{w} & -\overline{z} & 0 & d_+ \end{pmatrix}, \qquad a_\pm = \frac{1+c\pm t}{4}, \quad d_\pm = \frac{1-c\pm t}{4}. \tag{68}$$

Positivity implies $0 \leq t \leq 1-c$. Both spectral diagonalizations remain valid for repeated eigenvalues and singular density matrices. The trace-zero condition on C, which forces its diagonal entries to be $z, -z$, follows from the vanishing off-diagonal entry of Alice's marginal.

There is also a version with a prescribed radial axis. If Alice's Bloch vector is already ce_z, diagonalize only $H = A - D$. Direct Pauli expansion gives

$$\mathrm{BlochSq}(H) = \|T^{\mathsf{T}} e_z\|^2 = t^2. \tag{69}$$

Thus the prescribed t is preserved. This statement holds even when $c = 0$: the marked axis is still e_z, and Alice's unitary is exactly the identity. Diagonalizing a maximally mixed Alice marginal alone would not justify preservation of an arbitrary marked axis.

In the normal form,

$$\tau = t^2 + 16|z|^2 + 8|w|^2 + 8|v|^2. \tag{70}$$

The sharp geometric inequality $\tau \le 1 - c^2 + 2t^2$ has the following complete equality conditions:

$$tz = 0, \tag{71}$$
$$|z|^2 + |w|^2 = a_+ d_+, \tag{72}$$
$$|z|^2 + |v|^2 = a_- d_-, \tag{73}$$
$$z\overline{v} = w\overline{z}. \tag{74}$$

These are necessary and sufficient for every positive matrix (68). For $t > 0$, they reduce to

$$z = 0, \qquad |w|^2 = a_+ d_+, \qquad |v|^2 = a_- d_-. \tag{75}$$

Thus all extremizers in this basis are the explicit two-block states (75), including arbitrary admissible complex phases. At $t = 0$, the correct condition is instead

$$CC^\dagger = \frac{1 - c^2}{16} I. \tag{76}$$

In particular, setting $z = 0$ at $t = 0$ would discard legitimate extremizers. Equation (76) includes $c = 1$, when $C = 0$.

For necessity, the geometric spin-flip proof gives the exact identity

$$4 \operatorname{Re} \operatorname{tr}(M\widetilde{M}) = 1 - c^2 + 2t^2 - \tau.$$

Both M and $\widetilde{M}$ are positive. Zero trace therefore implies $M\widetilde{M} = 0$: conjugate by the positive square root of $\widetilde{M}$, use that a positive matrix with zero trace is zero, and then use the kernel of that square root. Four entries of the zero product give (71)–(74). Conversely, the two norm identities and (70) give equality immediately. This proof never divides by an eigenvalue or a diagonal entry.

A.2 Actual matrix stability for the canonical envelope

Put $d = 1 - c$, and define the canonical maximizing reference

$$M_{\mathrm{ob}}(c) = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & \sqrt{d} \\ 0 & c & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{d} & 0 & 0 & d \end{pmatrix}. \tag{77}$$

This is a Bob Pauli-X conjugate of the standard maximally obese state. For every canonical state σ, let

$$c = |a|, \qquad \delta = (1 - c)(3 - c) - \tau(\sigma) \ge 0.$$

If $0 \le c < 1$, there exist genuine qubit unitaries U, V such that, with $\sigma' = (U \otimes V)\sigma(U \otimes V)^\dagger$,

$$\|\sigma' - M_{\mathrm{ob}}(c)\|_1^2 \le 4\|\sigma' - M_{\mathrm{ob}}(c)\|_{\mathrm{HS}}^2 \le \delta \left(\frac{1}{2} + \frac{2 + c}{1 - c + t} \right) \le \frac{5 + c}{2(1 - c)} \delta, \tag{78}$$

where t is the radial parameter of the normal form chosen above. The trace-norm estimate can be capped by the diameter of the state space:

$$\|\sigma' - M_{\mathrm{ob}}(c)\|_1^2 \le \min \left\{ 4, \frac{5 + c}{2(1 - c)} \delta \right\}. \tag{79}$$

Consequently the usual trace distance is at most $\min\{1, \frac{1}{2}\sqrt{(5+c)\delta/[2(1-c)]}\}$.

Here is the matrix estimate behind the theorem. An Alice diagonal phase unitary makes w real and nonnegative; the zero-coherence case is handled by choosing phase one. Set $y = d - t$, $w_0 = \sqrt{d}/2$, $Z = |z|^2$, and $J = |v|^2$. Direct expansion of the actual entries gives

$$\|M - M_{\mathrm{ob}}(c)\|_{\mathrm{HS}}^2 = \frac{y^2}{4} + 4Z + 2J + 2(w - w_0)^2, \tag{80}$$

$$\delta = d^2 - t^2 + 8(w_0^2 - w^2) - 16Z - 8J. \tag{81}$$

Positive principal minors give

$$Z \le \frac{(1+c+t)y}{16}, \qquad J \le \frac{(1+c-t)y}{16}, \qquad 0 \le w \le w_0.$$

Using $(w - w_0)^2 \le w_0^2 - w^2$ yields

$$4\|M - M_{\mathrm{ob}}(c)\|_{\mathrm{HS}}^2 \le \delta + y(3 + 3c - t).$$

The geometric inequality gives $\delta \ge 2y(d + t)$. Substitution proves the middle estimate in (78); monotonicity in $t \ge 0$ gives the last. The first estimate follows from the spectral theorem and Cauchy–Schwarz for four eigenvalues. The cap follows from positivity and trace one of both density matrices. These spectral and matrix identities are proved in Lean, rather than supplied as assumptions to scalar algebra.

Setting $\delta = 0$ proves the complete canonical equality classification:

$$\tau(\sigma) = (1-c)(3-c) \quad \Longleftrightarrow \quad \exists U, V :\ (U \otimes V)\sigma(U \otimes V)^\dagger = M_{\mathrm{ob}}(c). \tag{82}$$

The endpoint $c = 1$ is proved separately: the normal form has $t = 0$, and positivity forces all three coherences to vanish. No estimate containing $1/(1-c)$ is used at that endpoint.

A.3 The complete global maximizing orbits

For $0 \le c < 2/3$, put

$$D(c) = 2 - 2c - c^2, \qquad r_*(c) = \frac{c(1-c)}{D(c)}, \qquad d = 1 - c.$$

Define the density matrix

$$\rho_*(c) = \frac{1}{2}\begin{pmatrix} 1 + r_* & 0 & 0 & \sqrt{d(1-r_*^2)} \\ 0 & c(1-r_*) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{d(1-r_*^2)} & 0 & 0 & d(1-r_*) \end{pmatrix}. \tag{83}$$

Every state with actual steering-centre length c satisfies

$$\tau(\rho) = G(c) \quad \Longleftrightarrow \quad \exists U, V :\ (U \otimes V)\rho(U \otimes V)^\dagger = \rho_*(c). \tag{84}$$

The theorem assumes no rank condition. If equality holds, $G(c) > 1$ forces Bob's marginal to be invertible, because a pure Bob marginal gives $\tau \le 1$. The global deficit identity then forces the canonical representative to saturate its canonical envelope. Apply (82) and conjugate the positive reconstruction filter by the same Bob unitary.

The remaining filter orientation is fixed by an exact identity. For the canonical reference (77) and any normalized complex filter with effect $F^\dagger F = I + b \cdot \sigma$,

$$G(c) - \tau(F M_{\mathrm{ob}}(c) F^\dagger) = (2 - 3c)(b_x^2 + b_y^2) + D(c)(b_z - r_*)^2. \tag{85}$$

The displayed notation means filtering on Bob. Both coefficients are strictly positive for $c < 2/3$, so every maximizing effect is exactly $I + r_* \sigma_z$. The reconstruction filter is positive, hence uniqueness of the positive square root gives $F = \mathrm{diag}(\sqrt{1+r_*}, \sqrt{1-r_*})$, which proves (83). The effect argument covers all complex normalized filters, including singular ones; no unproved polar decomposition is needed.

For $2/3 \le c < 1$, the supremum $G(c) = 1$ is not attained, as proved by the strict flat-branch bound for both marginal-rank cases. At $c = 1$, the only maximizers are pure product states. Indeed an invertible Bob marginal would give the strict joint bound $\tau < 1$; with singular Bob, $\tau = 1$ forces Alice's marginal to be pure too. Ordered qubit diagonalization puts both marginals at $|0\rangle\langle 0|$, and the positive support identity then forces the entire joint state to be $|00\rangle\langle 00|$.

A.4 Formal coverage of the matrix statements

The new modules use the same pinned Lean 4.19.0 and mathlib revision as the original development. The principal declarations are `canonical_aligned_normal_form`, `physical_bob_radial_normal_form`, `physical_radial_equality_iff`, `canonical_trace_rigidity_capped`, `canonical_obese_equality_iff`, `alignedObese_optimal_effect`, `global_steering_equality_iff`, and `global_endpoint_equality_iff`. Their hypotheses are positivity, normalization, and the stated measured coordinates; actual unitary witnesses, matrix distances, and singular support are derived inside Lean. The marked-axis theorem uses e_z explicitly, including at zero centre. The global maximizing-orbit classification has no coordinate or local-unitary lifting hypothesis.

B Relation to the earlier package

The earlier package [2] contains the envelope (14), its witnesses, the nonattainment on the flat branch and the fully entangled fraction bound. Its proof was read again for this paper; the table records the status of each step.

Step	Finding
Canonical spectral estimate	Valid. For a top eigenvector of weight λ, the two pure-state marginal Bloch lengths agree. Positivity of the Bob remainder gives $\lambda r_0 \le 1-\lambda$; the Alice triangle inequality gives $c \le \lambda r_0 + 1 - \lambda \le 2(1-\lambda)$. Thus $\lambda_{\max} \le 1 - c/2$. This also follows from established marginal constraints [7]; it is not a novelty claim.
Purity bound	Valid. A probability vector with each entry at most $L = 1 - c/2 \ge 1/2$ has squared sum at most $L^2 + (1-L)^2$. This gives $\mathrm{Tr}\,\sigma^2 \le 1 - c + c^2/2$ and the canonical correlation bound.
Filter covariance	Valid as a polynomial identity. It remains valid for singular filters. The earlier manuscript uses FXF for Hermitian physical filters; it does not replace $F^\dagger$ by F for arbitrary physical filters.
Marginal estimate	Valid by positivity of the weighted partial trace. The proof does not assume that filtering preserves F_3.
Scalar maximization	Exact; the completed square and flat-branch deficit identities verify the rational branch and the exact threshold.
Degenerate cases	Singular global states are correctly separated from singular Bob marginals. The point-ellipsoid convention is essential.
Maximizers	The earlier package gives sharp witnesses and nonattainment, but not a proof that all curved-branch maximizers belong to the displayed local-unitary orbit. Theorem 13 supplies that classification.

Step	Finding
Formal scope	The formal target is the actual positive semidefinite matrix correlation quantity and the computed canonical centre. It does not formalize measurement models or EPR steering in general.
Fully entangled fraction	Not a result of this paper; the Omega Institute's public repository records a claimed Lean proof of the same bound [12], with a related CHSH record [13].

No counterexample or invalid inference was found in the earlier envelope proof. A successful Lean kernel check establishes the formal declarations, not the accuracy of every informal phrase in an accompanying text.

C Formal and computational verification

C.1 The Lean project and its audit

The Lean project in `lean/` pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`, with every transitive dependency fixed in its manifest. It has 70 modules in `lean/Lean/`, listed in dependency order in `lean/modules.txt`. Eighteen of them are the original modules of the earlier package [2], in the form in which they were first produced (the earlier package's copies differ only by linter clean-ups, docstring wording and added `#check` commands); the other 52 are new. The audit module `MilneAxiomAudit` enumerates every declaration whose defining module is one of the 70, including private and compiler-generated ones, and collects its transitive axioms. Apart from the code-generation auxiliaries that Lean 4.19 stores for executable definitions, which may refer only to the compiler placeholder `lcProof` and to each other, every declaration is a non-axiom depending only on `propext`, `Classical.choice` and `Quot.sound`. The sources contain no `sorry`, `admit`, `axiom` declaration, `native_decide` or `ofReduceBool`. The fresh build and audit recorded with this package compiled all 70 modules and audited 1,298 declarations, 1,044 of them other than code-generation auxiliaries; six modules only print axiom reports and define no declarations.

`Density M` means that the complex matrix $M \in \mathbb{C}^{4\times4}$, indexed by `Fin 2 × Fin 2`, is positive semidefinite with trace one. `correlation M` is the real matrix of Pauli expectation values $\mathrm{Tr}(M\,\sigma_i \otimes \sigma_j)$ and `correlationSq M` the sum of their squares. `steeringCenter` is computed by the positive canonicalizing filter $(2\rho_B)^{-1/2}$ when Bob's marginal is invertible and is Alice's Bloch length otherwise. `hermitianTraceNorm` is the sum of the absolute eigenvalues; the usual trace distance is half of it. Local unitaries are elements of the unitary group of $\mathbb{C}^{2\times2}$, and Kraus operators are complex 2×2 matrices G with $\mathbb{I} - G^\dagger G \succeq 0$.

C.2 Formal results and their domains

Mathematical result	Formal coverage
Global envelope and exact 2/3 threshold (earlier package)	Actual complex 4×4 positive semidefinite trace-one matrices; computed centre, least upper bound, sharp witnesses, attainment and nonattainment, and pure Bob marginals.
Centre-coordinate bridge	The whitening centre equals $(\mathbf{a} - T\mathbf{b})/(1 - \|\mathbf{b}\|^2)$ whenever Bob is nonsingular; the envelope and threshold are also stated in measured Pauli coordinates.

Mathematical result	Formal coverage
Global extremizer classification	Actual local qubit unitaries put every curved-branch maximizer into the displayed optimal family. The optimal complex filter effect is unique; no full-rank premise is imposed. At $c = 1$, the maximizers are exactly the pure-product unitary orbit.
Complete joint (c, r) interval	Equality of the actual density-matrix correlation spectrum with the closed interval in Theorem 5, for $r < 1$; a separate endpoint theorem gives c^2 at $r = 1$. Explicit physical interpolating states are supplied.
Lower-endpoint equality and stability	Exact matrix-coordinate equality conditions, correlation and marginal displacement bounds, spectral trace-norm stability, and a proof that the rotated reference is itself positive and trace one, including vanishing correlations.
Geometric matrix inequality	Reconstruction from arbitrary density matrices, the directional spin-flip trace identity, and the full real 3×3 positive semidefinite conclusion.
Complete canonical (c, t) interval	Exact attained interval, its feasible range, explicit physical witnesses, and lower-endpoint equality. The upper endpoint has a complete algebraic equality criterion after an explicit Bob unitary with the marked Alice axis fixed.
Canonical equality and trace-norm rigidity	Actual two-qubit local unitaries, complete equality with the canonical obese state, Hilbert–Schmidt and spectral trace-norm estimates, and the norm-diameter cap, including the zero-coherence and pure-marginal cases.
Probability-constrained filtering	Exact optimum and activation criterion for actual contracting complex Kraus operators and actual normalized output states; the success trace and radius restriction are derived. The $p = 1/2$ boundary and all looser budgets are included.
Complete one-sided orbit	Every normalized complex filter, singular or invertible. The unique admissible spectral shift exists, lies among at most seven algebraic candidates, and gives the exact maximum and every maximizing effect. The open orbit has the same supremum, with an exact attainment criterion.
Volume–correlation geometry	Exact signed bounds, rank-safe row tests, physical sharpness, signed-determinant invariance under local unitaries, and the attained three-parameter volume maximum over actual density matrices in the fixed marked-axis convention below.

At $c = 0$, a specified radial direction remains part of the data. The volume formalization takes that marked Alice direction to be $\mathbf{e}_z$: its domain is all density matrices with Alice Bloch vector $c\,\mathbf{e}_z$, Bob Bloch vector zero and $\left\| S^{\mathsf{T}}\mathbf{e}_z \right\| = t$, and it proves the needed Bob-only normal form. The rotation to an arbitrary marked frame is a change of coordinates in the written proof. The canonical and global local-unitary classifications have no marked-axis premise.

C.3 Principal declarations

- `Milne.steeringEnvelope_isLUB`, `Milne.three_setting_violation_possible_iff`, `Milne.steeringCenter_coordinates`.

- `Milne.global_steering_equality_iff`, `Milne.global_endpoint_equality_iff`, `Milne.canonical_obese_equality_iff`, `Milne.canonical_trace_rigidity_capped`.
- `Milne.jointSteeringSpectrum_eq_Icc`, `Milne.actual_joint_value_iff`, `Milne.joint_lower_equality_iff`, `Milne.joint_physical_trace_rigidity`, `Milne.arbitrary_joint_trace_rigidity`.
- `Milne.geometryPhysicalSkewSpectrum_eq_interval`, `Milne.physical_radial_equality_iff`.
- `Milne.successPhysicalSpectrum_isGreatest`, `Milne.successPhysicalSpectrum_violation_iff`.
- `Milne.trust_global_optimum_iff_certificate`, `Milne.canonical_filter_finite_spectral_solution`, `Milne.canonical_filter_all_maximizers`, `Milne.canonical_invertible_F3_attainment_iff`.
- `Milne.physicalTripleVolumeSpectrum_isGreatest`, `Milne.physicalRadialVolumeSpectrum_isGreatest`, `Milne.localConjugate_correlation_det`.

C.4 Statements with conventional proofs

The sharp flat-branch distance exponents in Theorem 16, the rank counts of the interpolation and volume families, the phase-gauge simplifications of the fixed-radial equality equations, the attainment of every value in the signed-determinant interval, and the full piecewise classification of all volume maximizers have conventional proofs only. Lean checks the physical bounds and sharp witnesses, including their singular endpoints, but does not separately package every rank count or phase-gauge corollary. The ellipsoid-only and adjugate restatements and the composition of an optimal canonical filter with a preceding whitening operation are likewise written consequences.

The identification of $\|T\|_{\mathrm{F}}$ with the optimized three-setting CJWR expression [6], the steering-ellipsoid interpretation of the canonical centre, local-hidden-state models and steering under arbitrary measurements are outside the formal development. The success frontier prices a filter applied to a canonical input; it does not include the cost of a preceding whitening operation.

C.5 Exact and numerical checks

The checks in `calculations/` are reproducible and record their seeds and residuals; they support debugging and corroborate the proofs, and no universal statement rests on them. `calculations/reproduce_v2.py` (seed 2406039452026) checks 2,000 random density matrices, 500 of each rank, 160 joint-interval witnesses, canonical radial interpolations, 480 sharp volume witnesses, physical Kraus operations and 126 quadratic problems with regular, singular, indefinite and hard-case data; the largest optimizer discrepancy was 2.4×10^{-11} and the largest apparent violation of a volume bound 8.2×10^{-14}, at roundoff scale. `calculations/reproduce.py` (seed 240603945) checks 20,000 random states against the filter identity, the global bound, the matrix inequality, the deficit identity and the stability estimate. `calculations/exact_orbit_certificate.py` and `calculations/exact_filter_certificate.py` verify six exact rational certificates each, using all principal minors of the physical state and of the shifted quadratic matrix; they reject floating-point input. `calculations/exact_physical_example.py` verifies the activation example of Section 6 in rational arithmetic. The scripts in `calculations/independent/` are separate implementations of the same checks, and `calculations/independent/exact_new_results.py`, written for this release, verifies the witnesses and scalar identities of the joint interval, the volume trade-off and the probability frontier in exact symbolic arithmetic. All of these were rerun for this release and reproduced the recorded outputs. The Python programs are not formally verified.

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